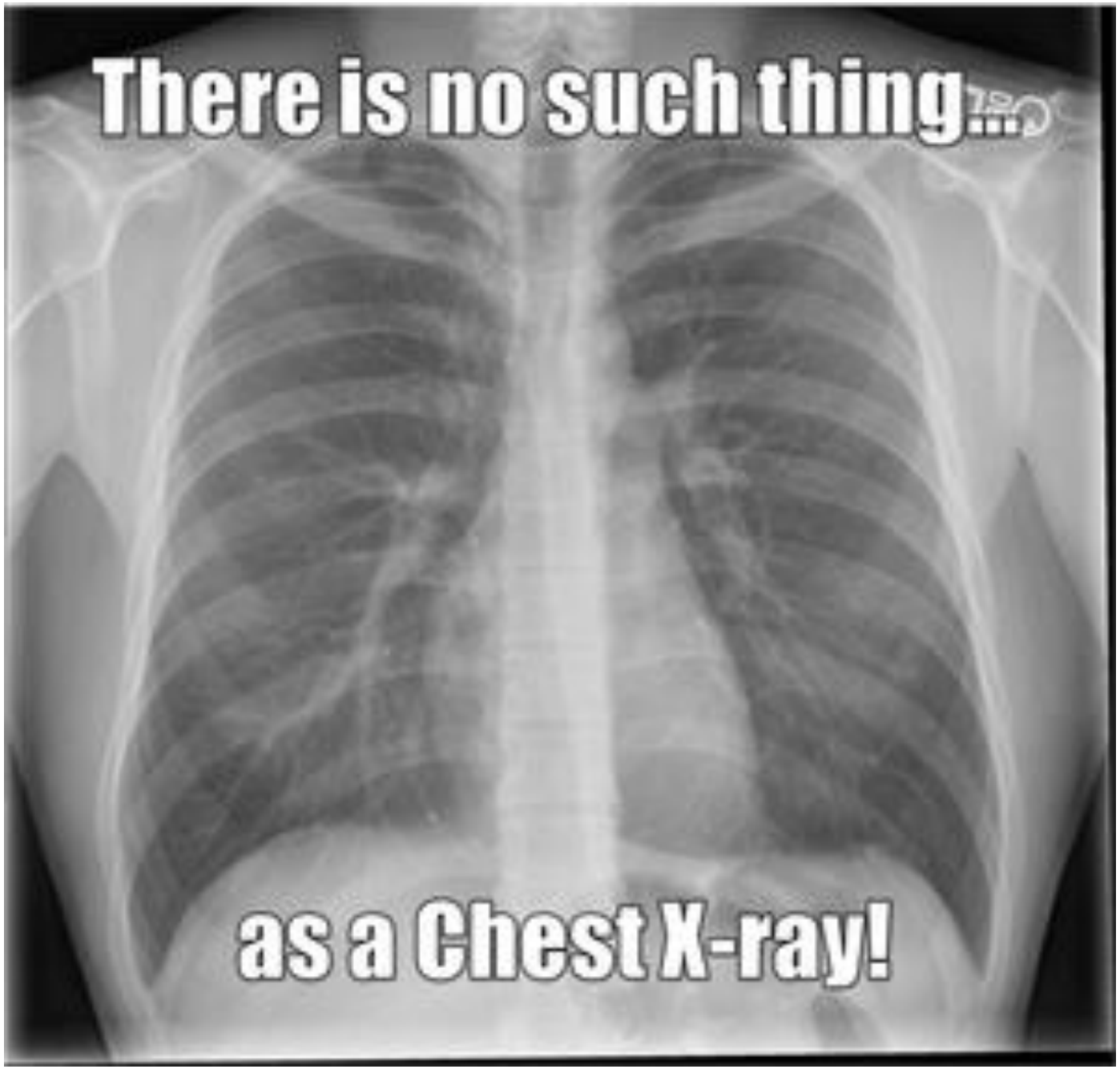
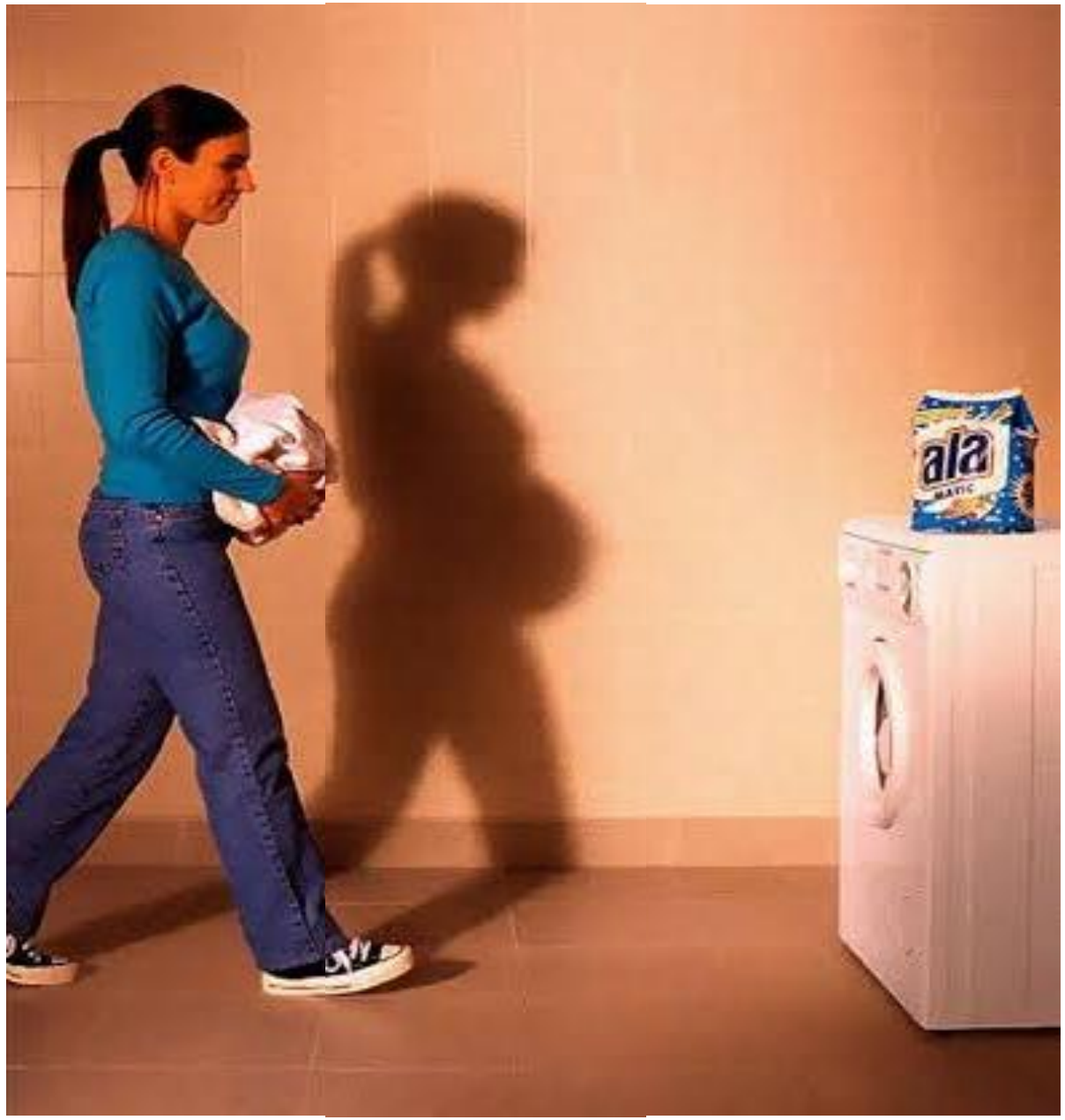
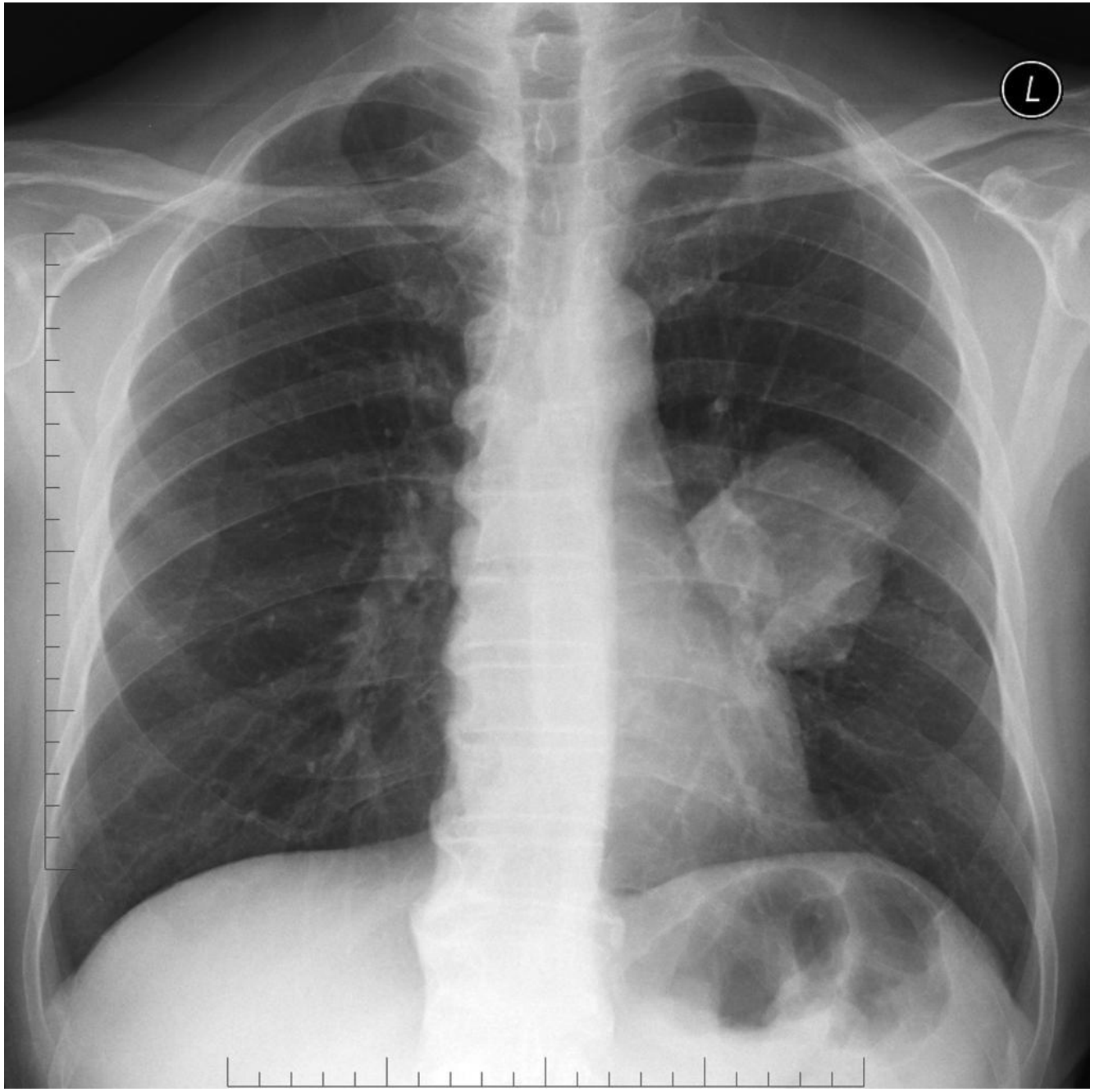
Reading a Chest Radiograph
Back to Basics

Maj (Dr) Joish Upendra Kumar
Assistant Professor
Radiodiagnosis
KVG Medical College and Hospital


There is no such thing...
as a Chest X-ray!

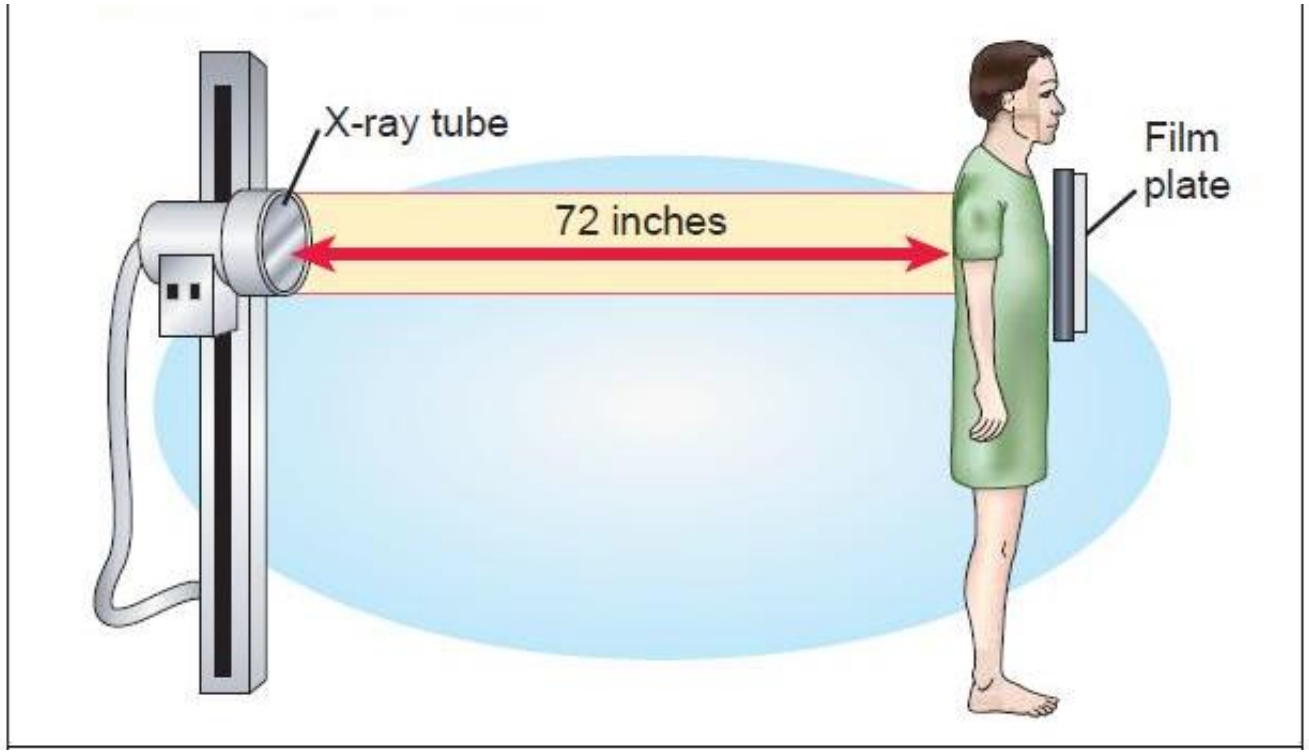
- It’s Chest Radiograph


ala
MATIC


L

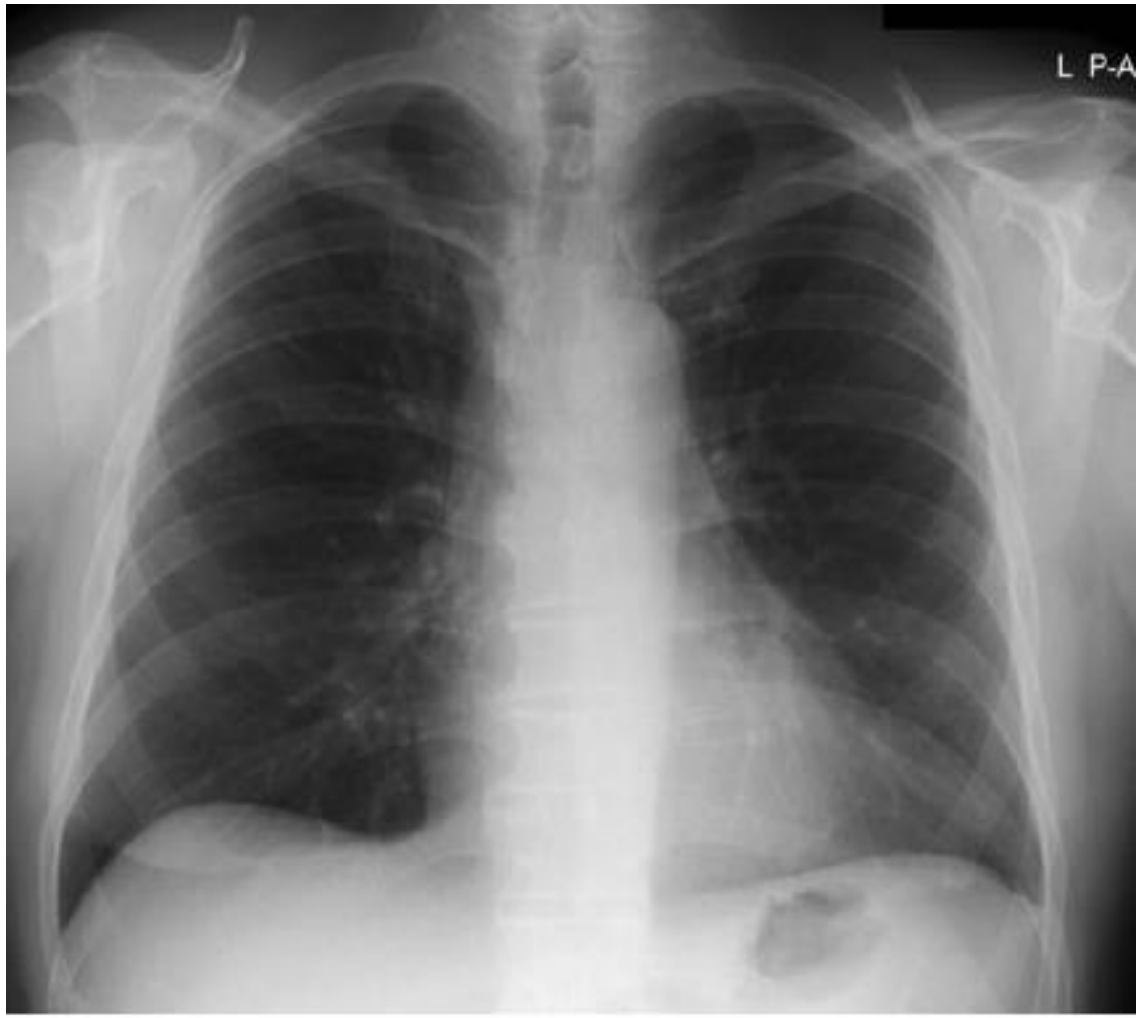

R


Right


X-ray tube
72 inches
Film
plate



PA vs AP



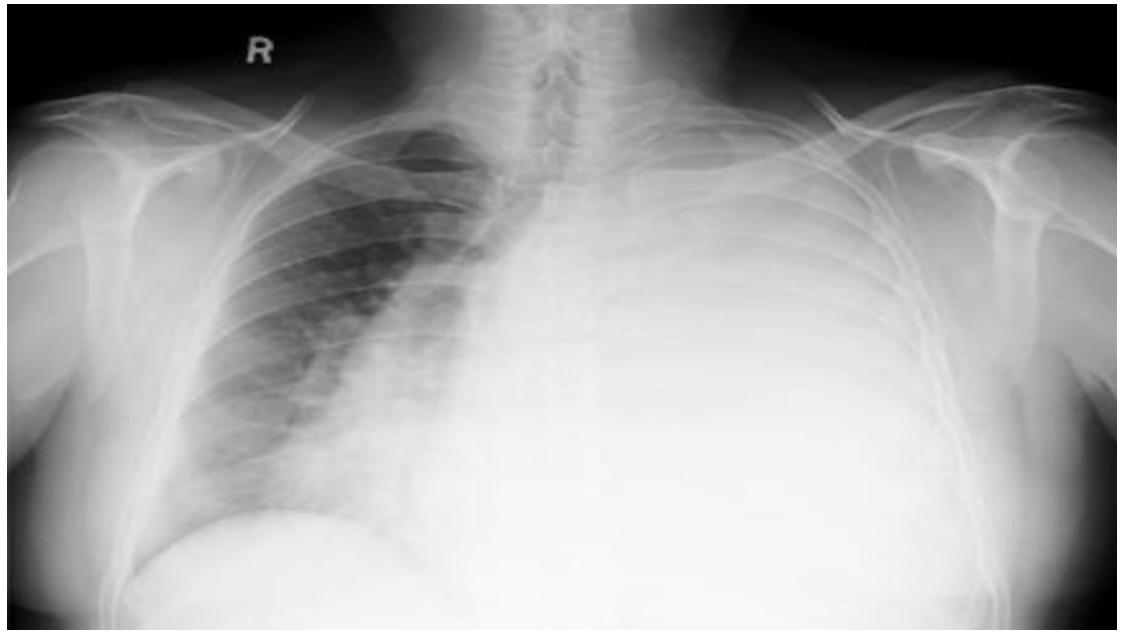


Exposure factors





Tracheal displacement



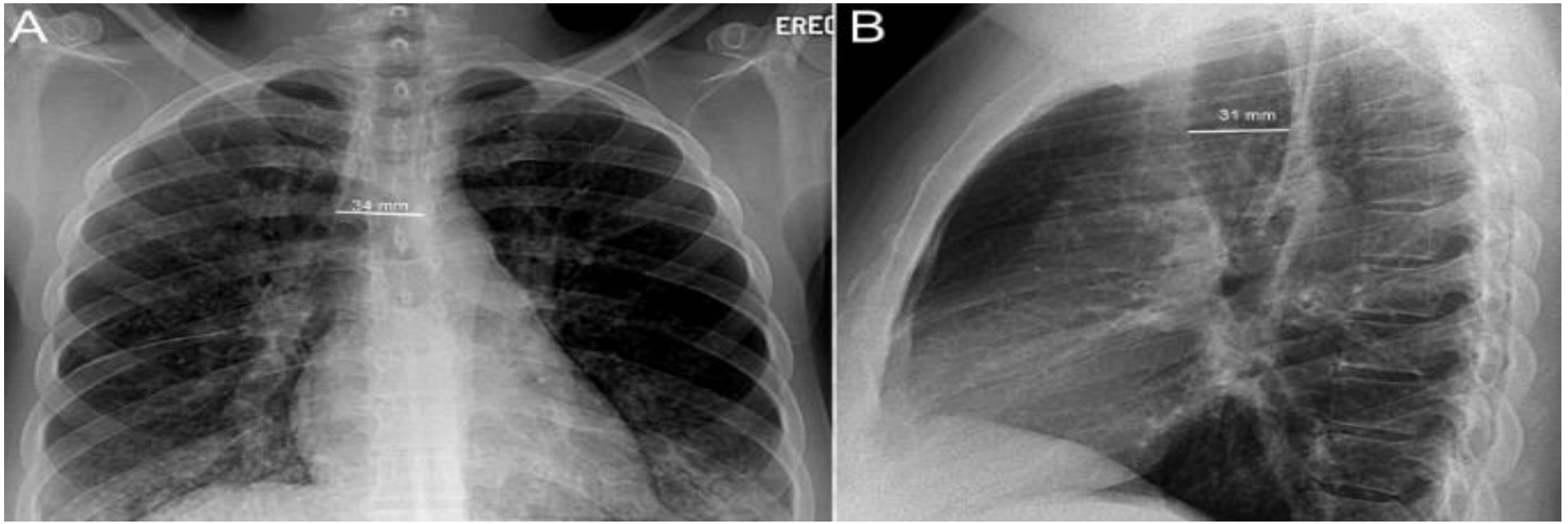


Tracheal narrowing



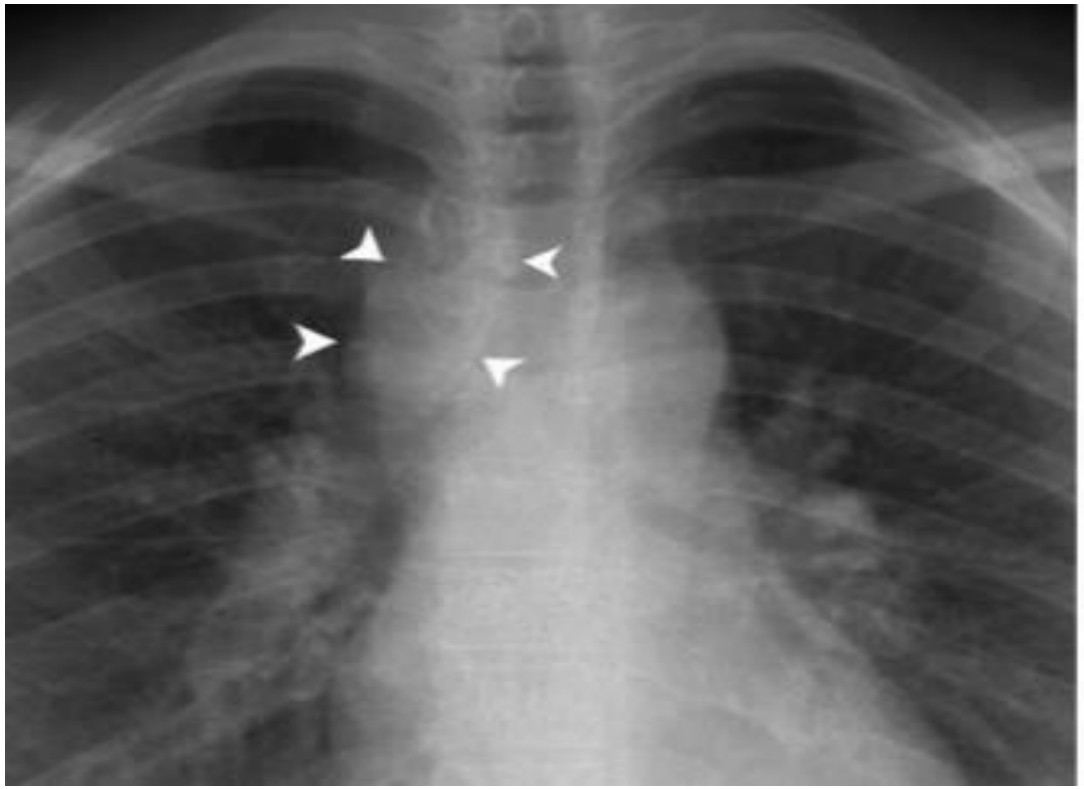
- Croup.
- Endotracheal mass.
- Foreign body.

Widening of trachea



- > 25mm seen in Tracheobronchomalacia (MKS)

Widening of paratracheal stripe

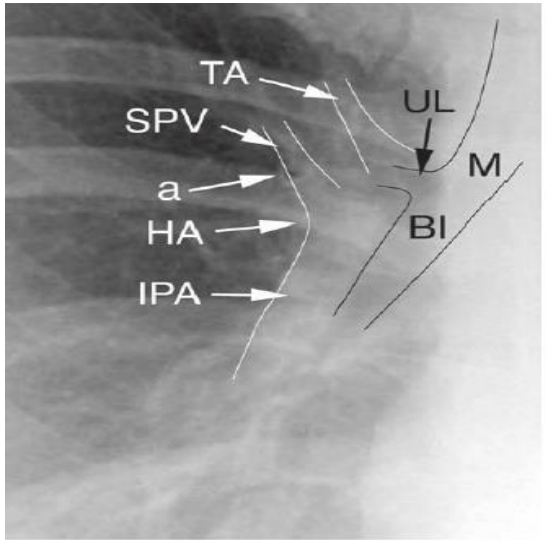


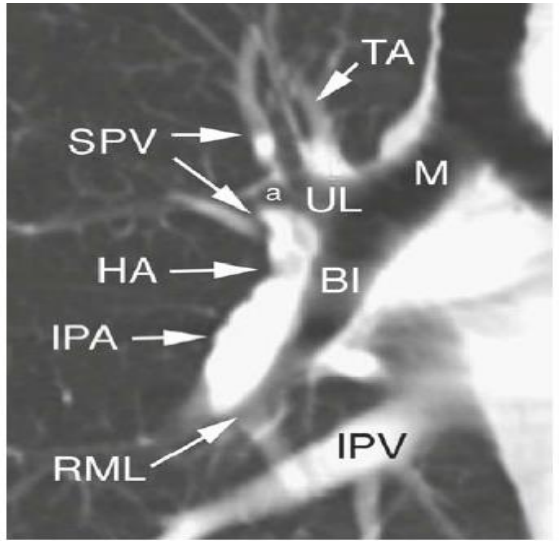
- Right tracheal para stripe widening
- Normal Rt = < 5mm, Lt = <10mm.
- LNs
- Tracheal wall tumors.
- Pleural effusion / pleural masses.

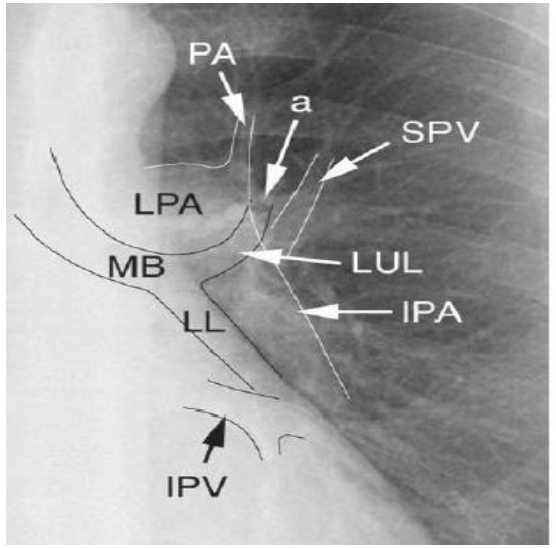


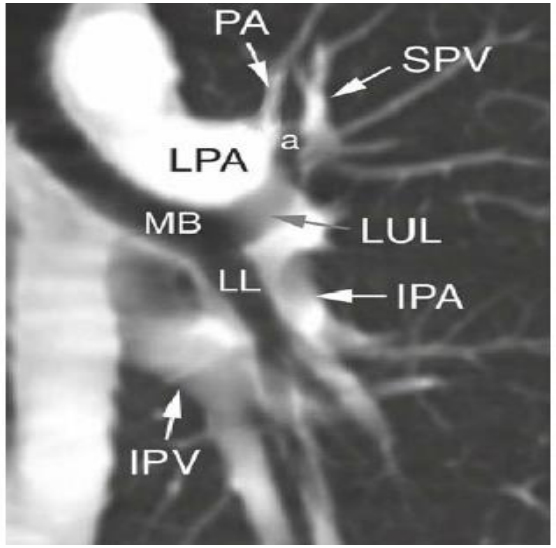
Hilum.

Position
Size & density.



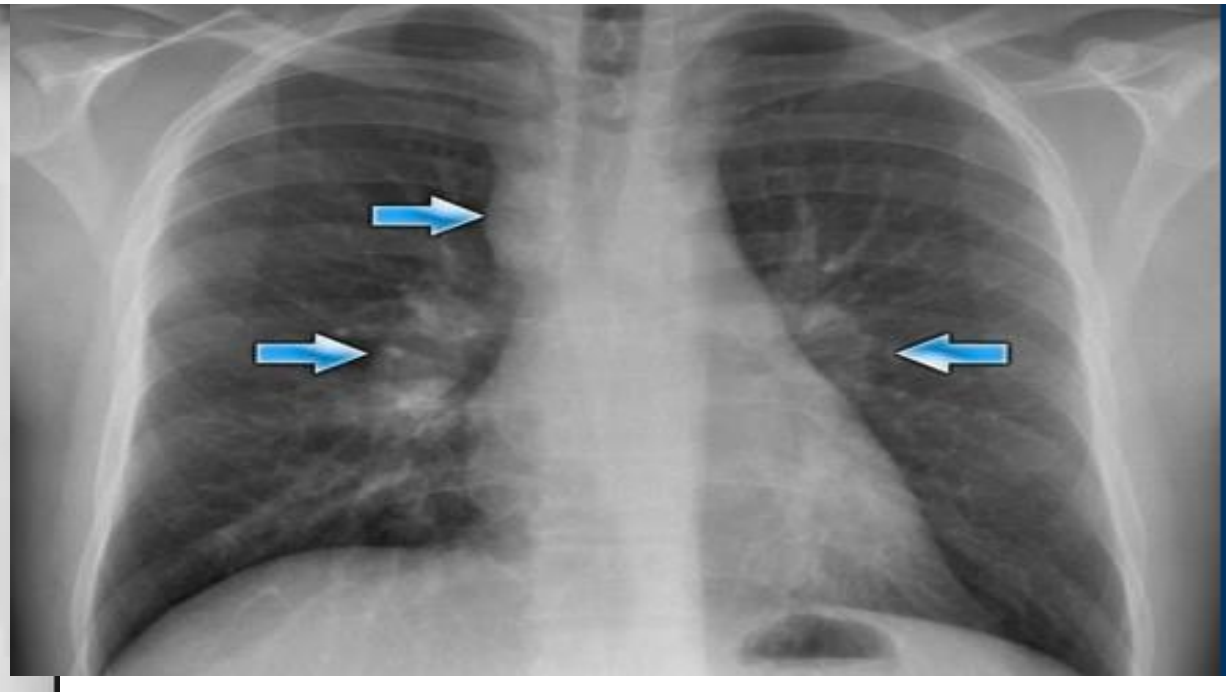


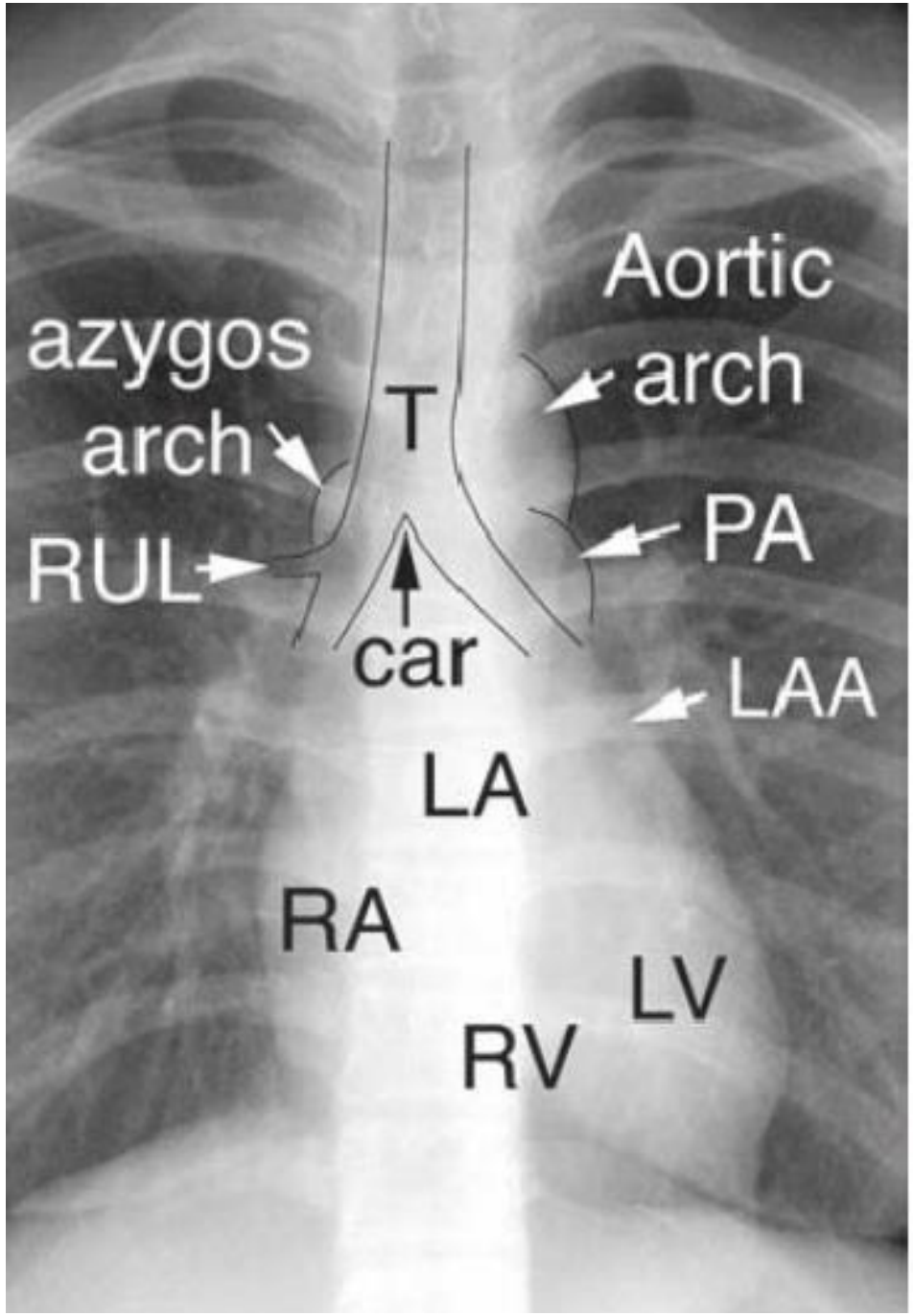


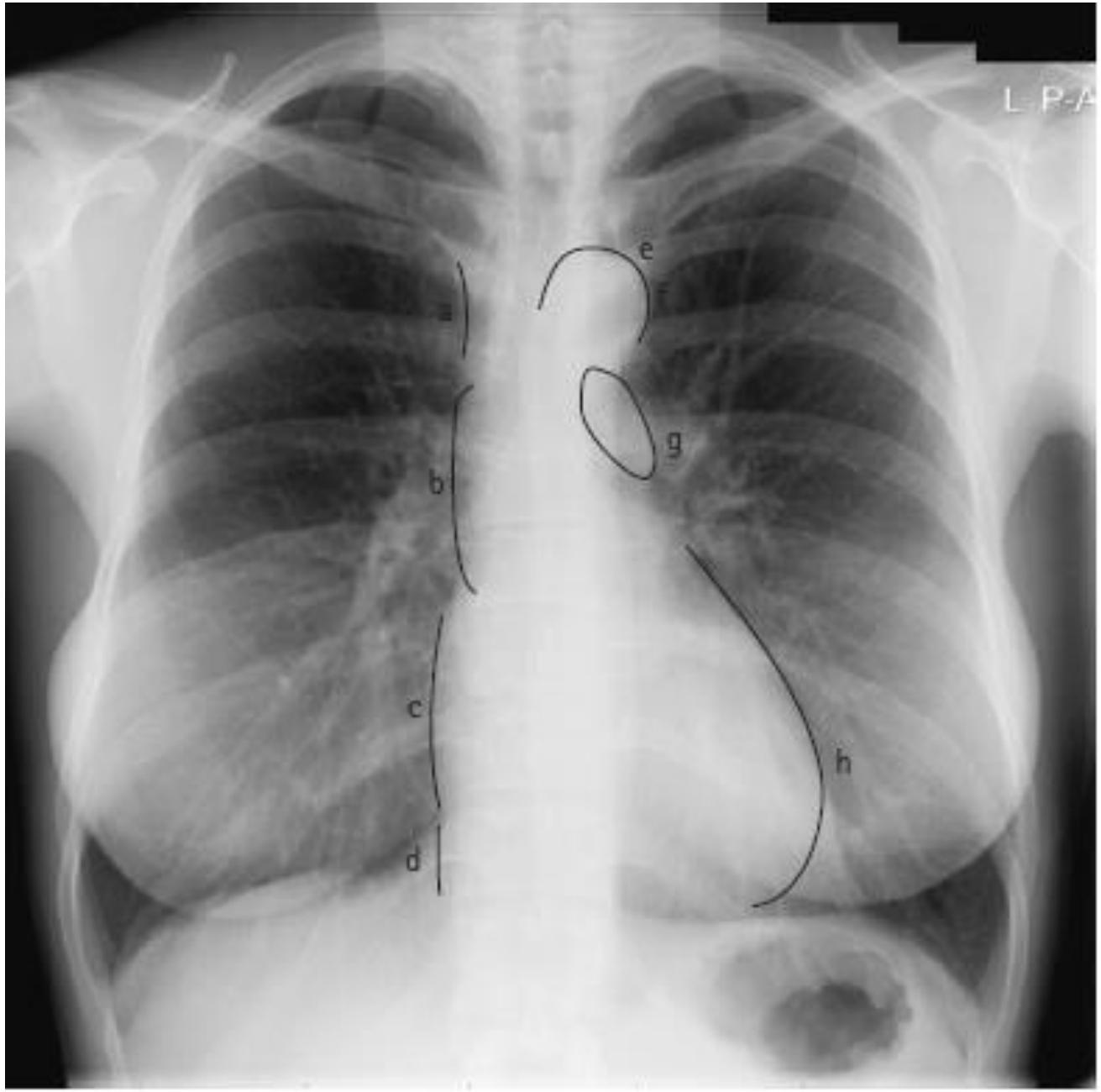


Enlarged and dense Hila

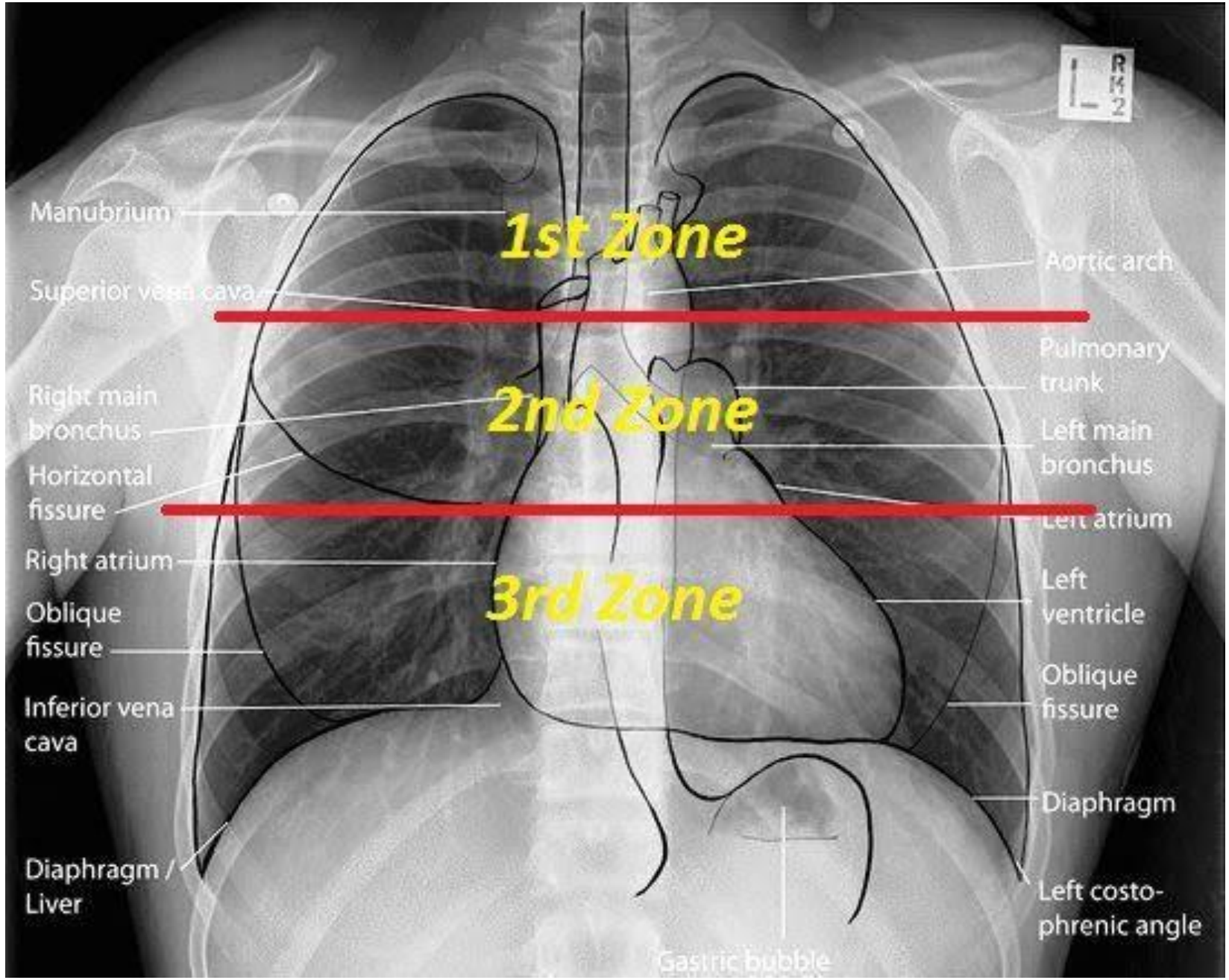




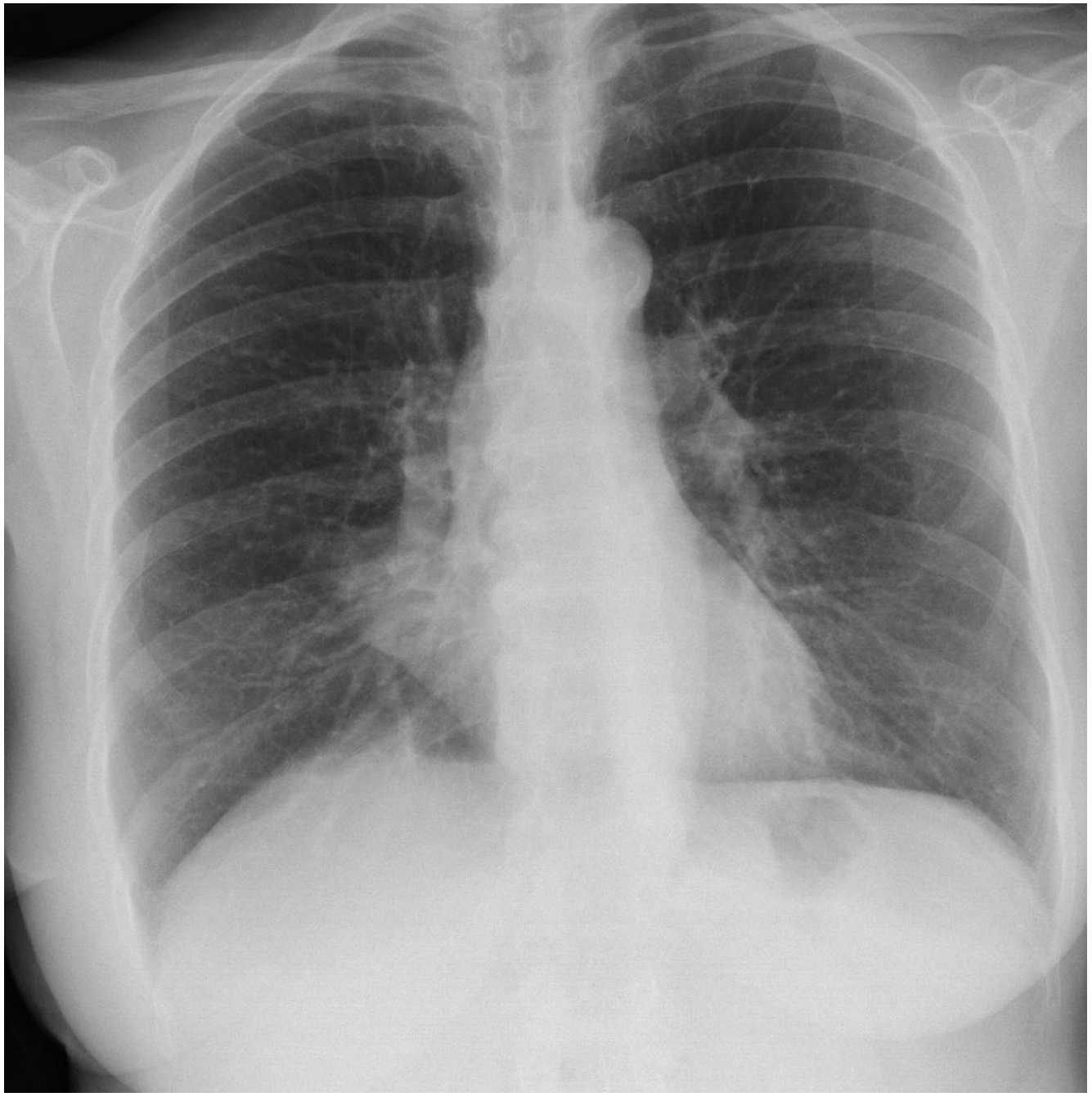


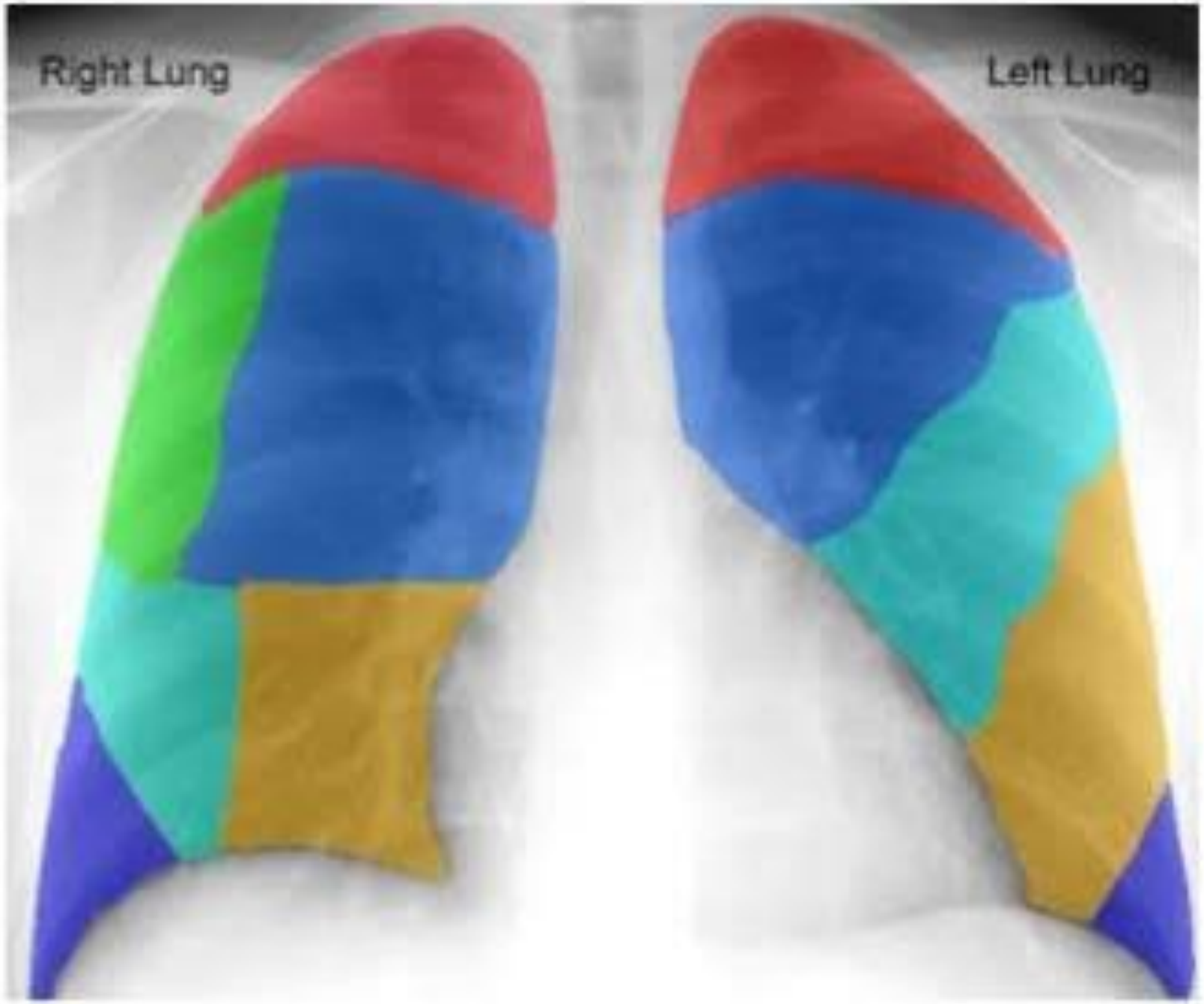


Ref – Chest Imaging - Webb


Manubrium
1st Zone
Superior vena cava
Aortic arch
Pulmonary trunk
Right main bronchus
2nd Zone
Left main bronchus
Horizontal fissure
Left atrium
Right atrium
3rd Zone
Left ventricle
Oblique fissure
Oblique fissure
Inferior vena cava
Diaphragm
Diaphragm / Liver
Left costo-phrenic angle
Gastric bubble

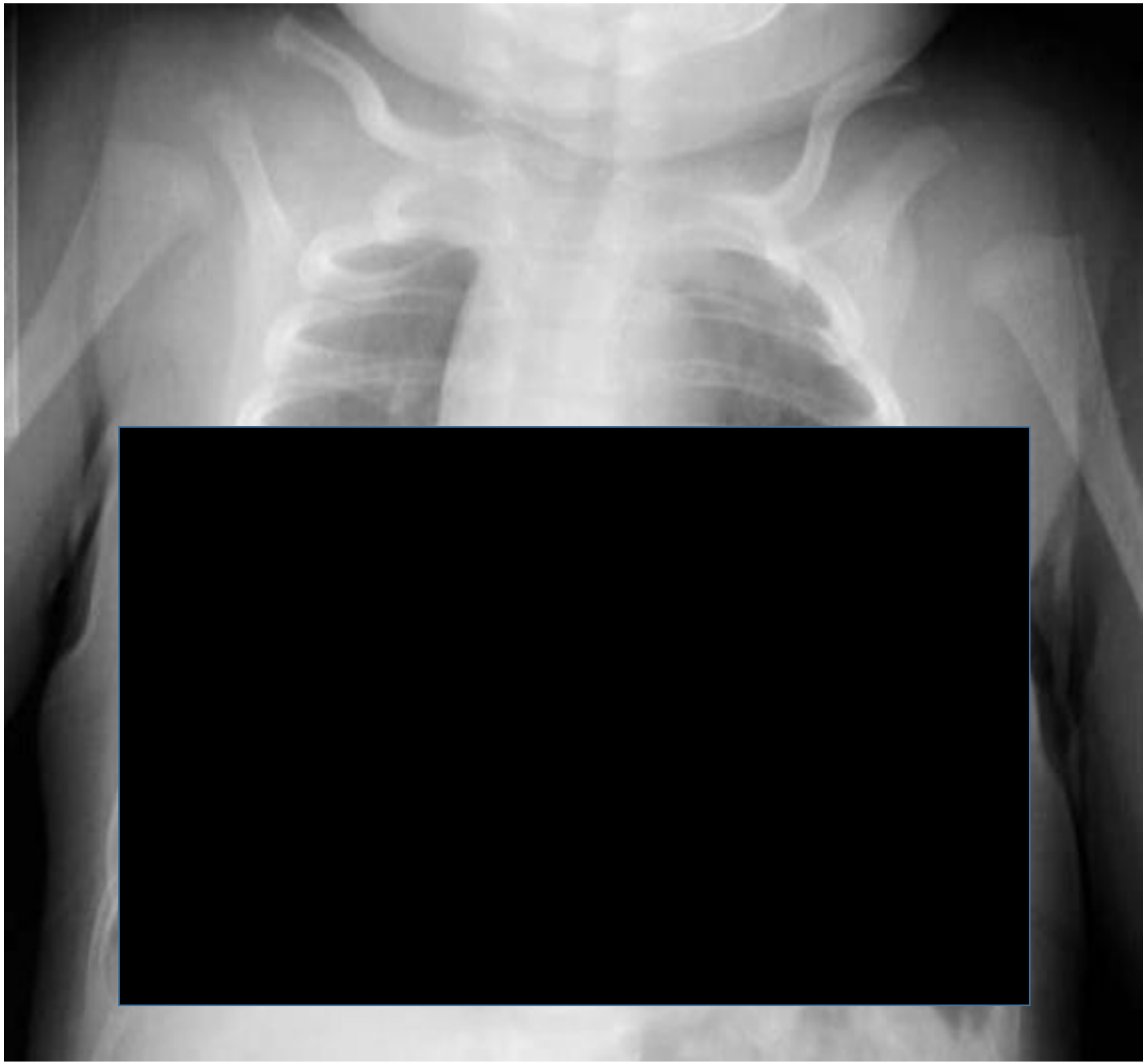
Silhouette sign



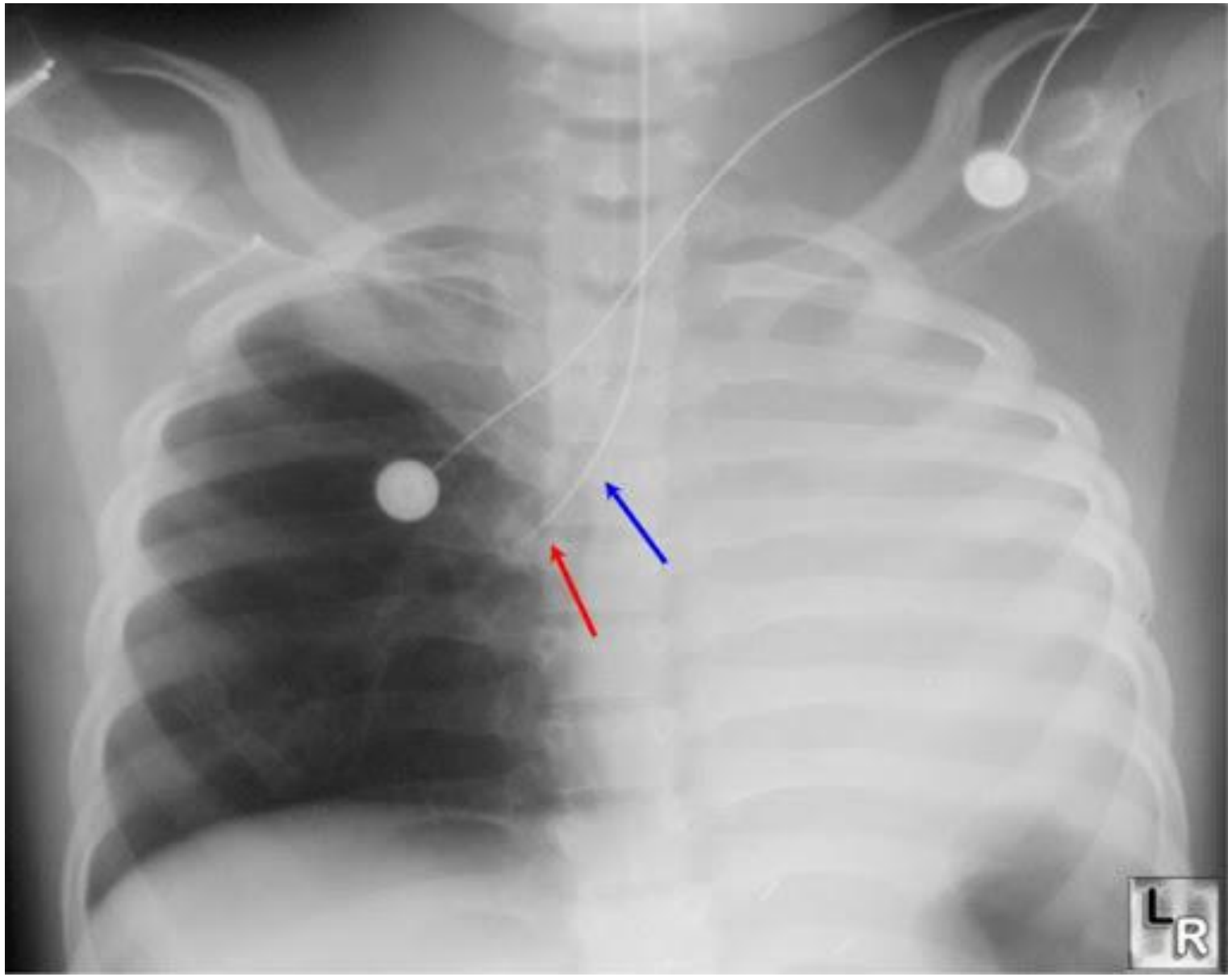

Right Lung
Left Lung

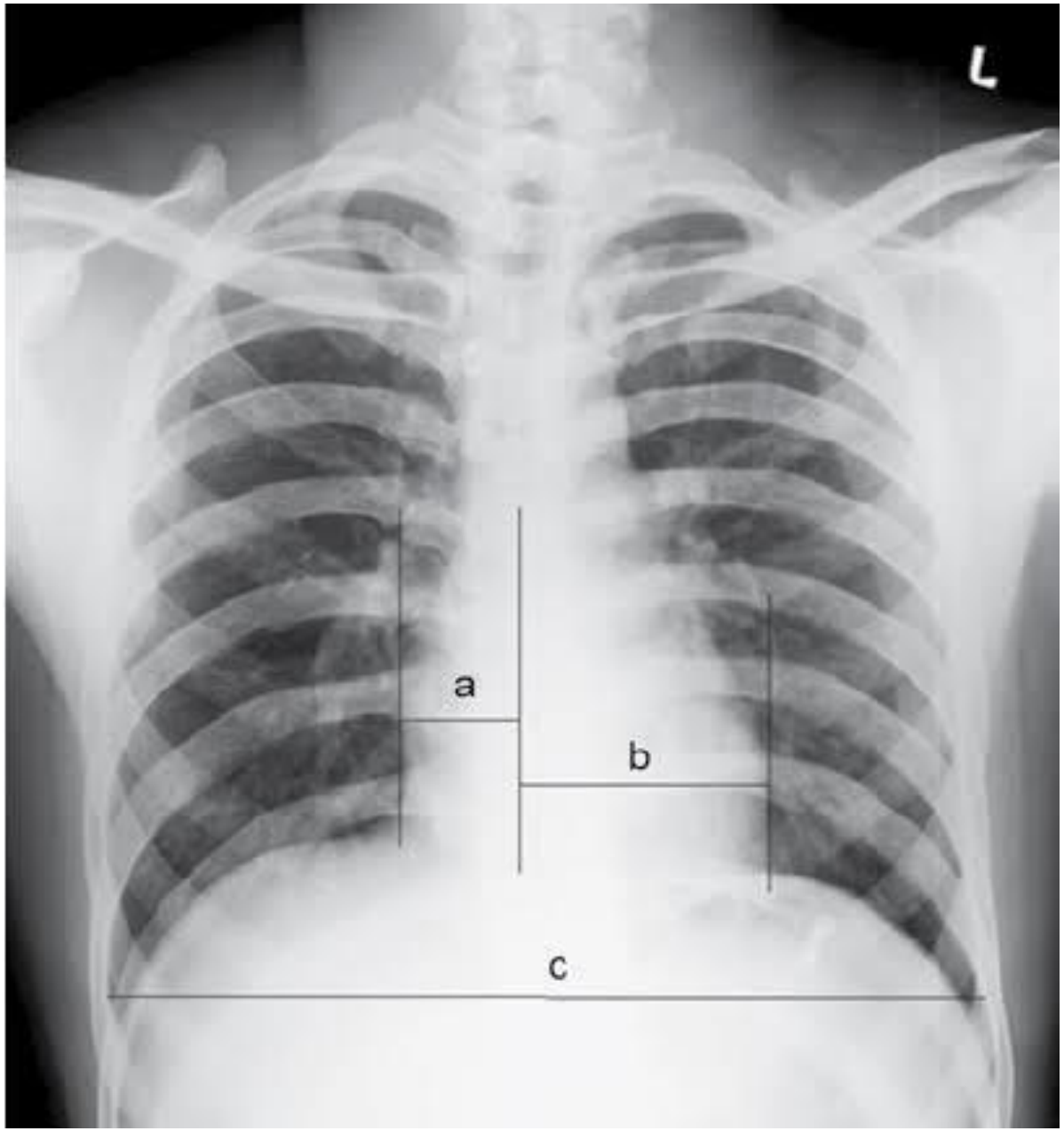


CONGENITAL LOBAR EMPHYSEMA

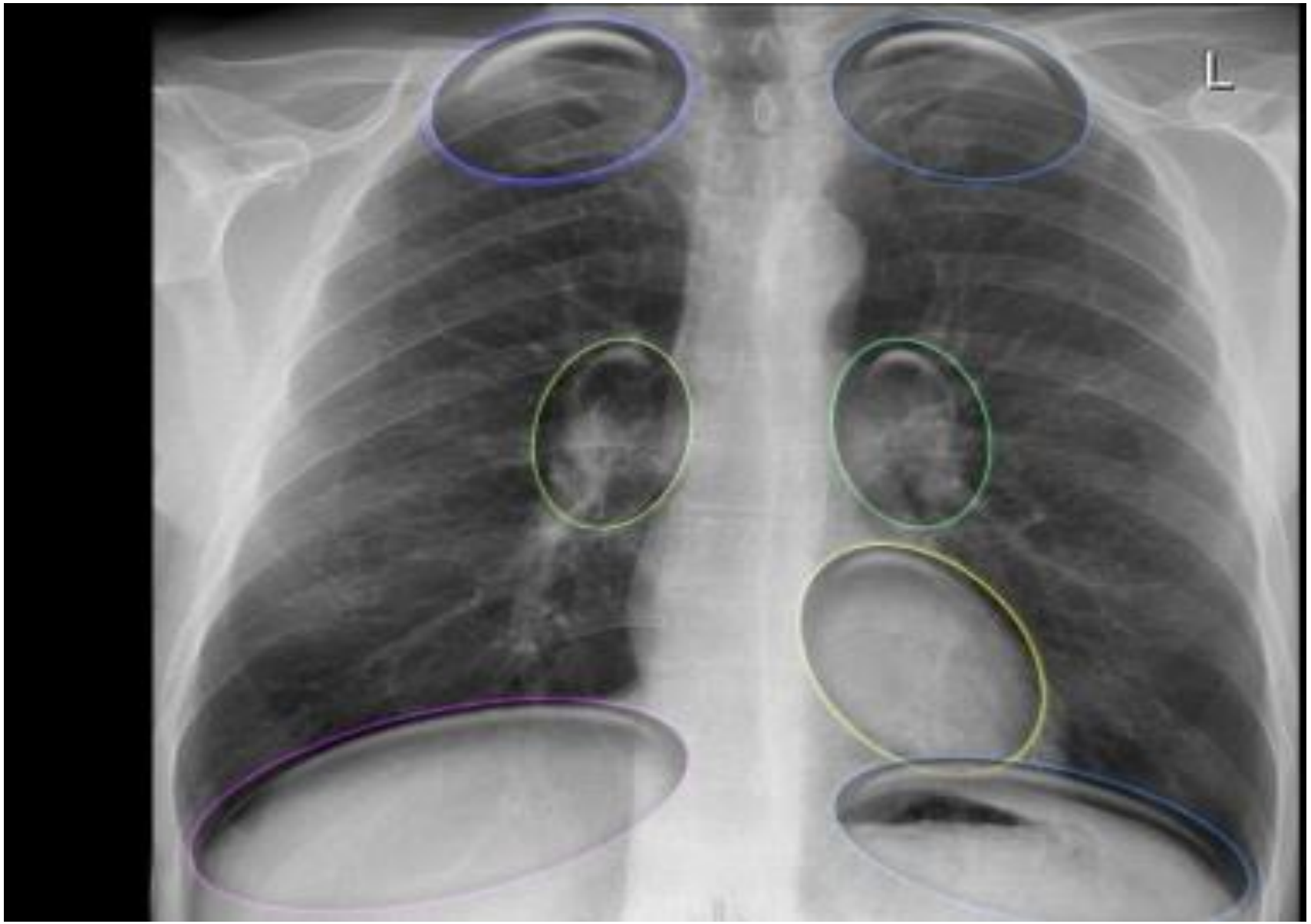





L
R


L
a
b
c

Hidden areas



- Apices.
- Hila
- Retrocardiac.
- Sub diaphragmatic.

*Radiologyassistant.nl


R

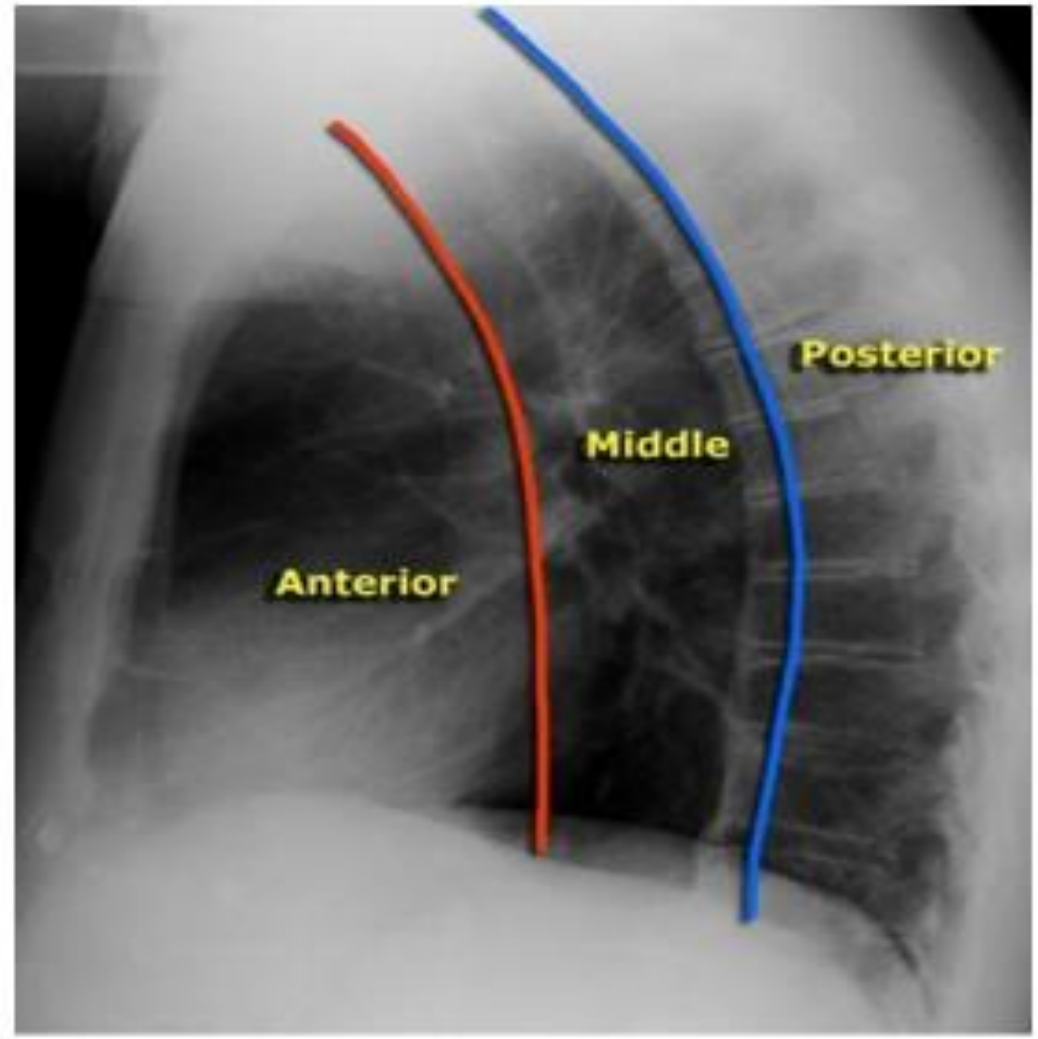

R
L
Zoom: 218% Angles L-R: 0°, S-I: -90°
Im: 99/167 (S → I) Series: 5
JPEG2000LosslessOnly
Thickness: 1.50 mm Location: -965.10 mm
P
28/11/19 10:02:44 am
Made In OsiriX

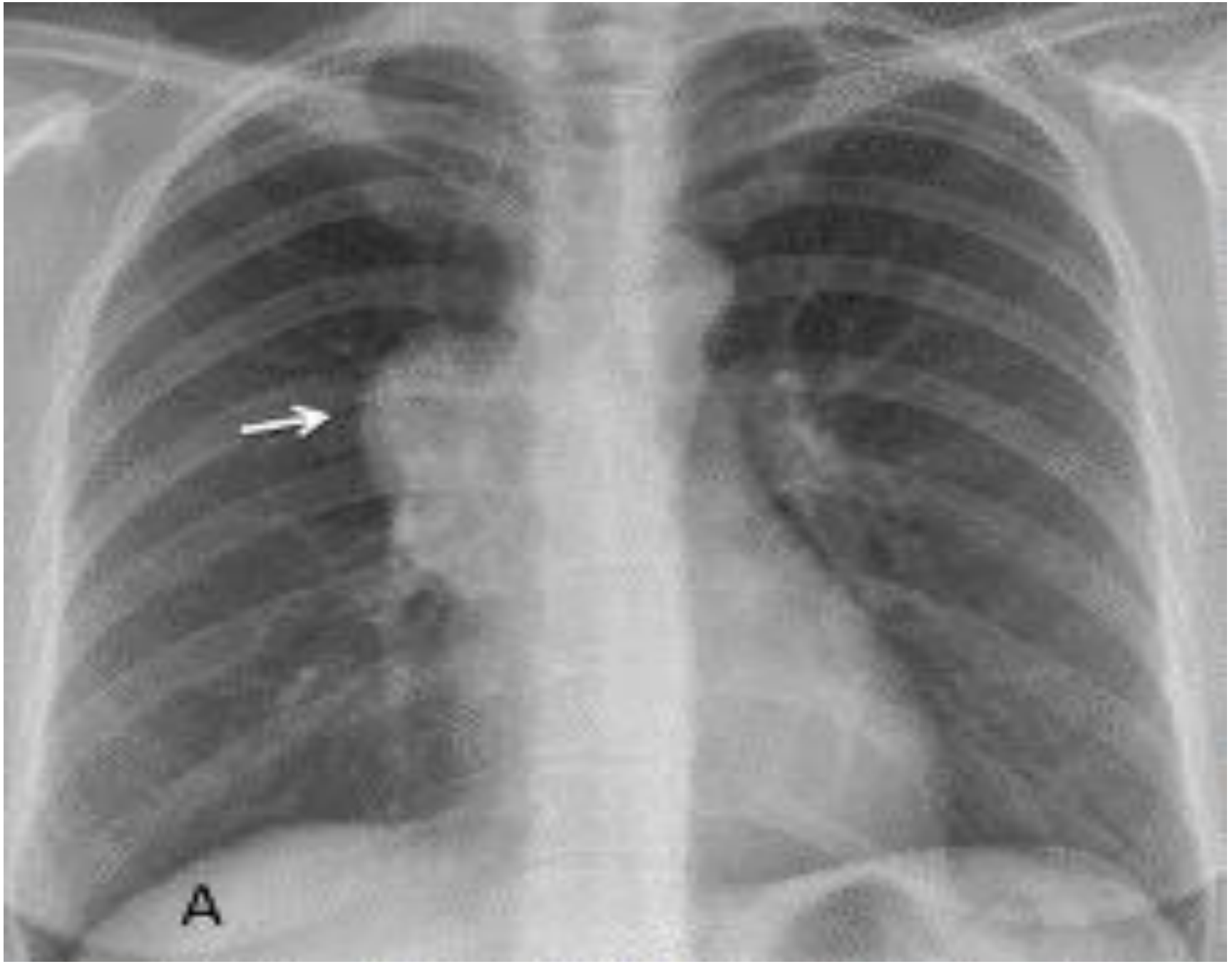
Mediastinal masses

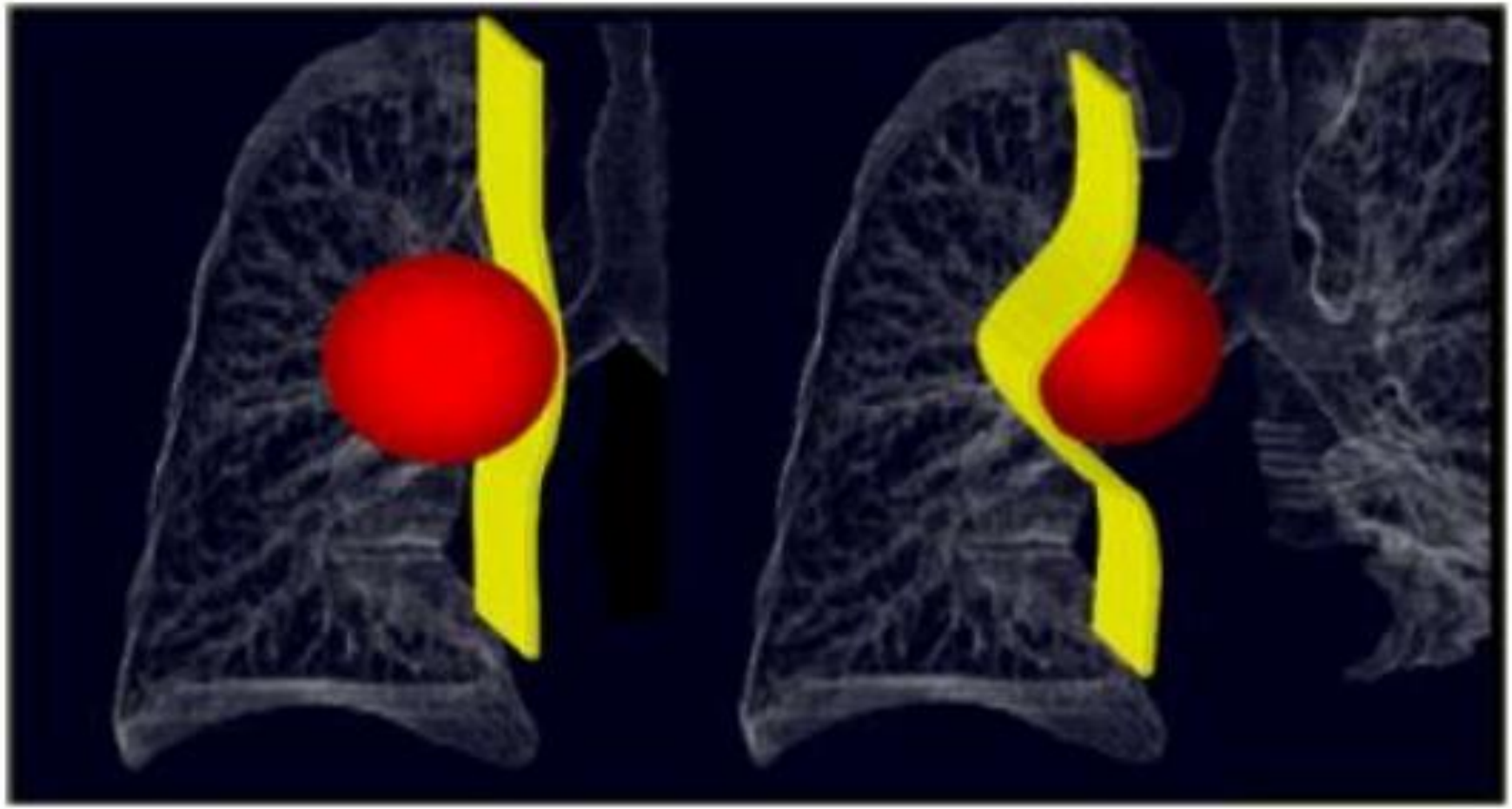
- Broad base – obtuse angle with lung
- Irregularity or lung infiltrates – lung lesion
- Smooth contour
- No bronchograms
- May be bilateral
- Hilum – Overlay, convergence
- Cervicothoracic/Abdominothoracic signs
- Silhouette sign – lung masses have sharp margins
- Mediastinal lines and stripes

Felson division of mediastinum

- Based on the lateral CXR



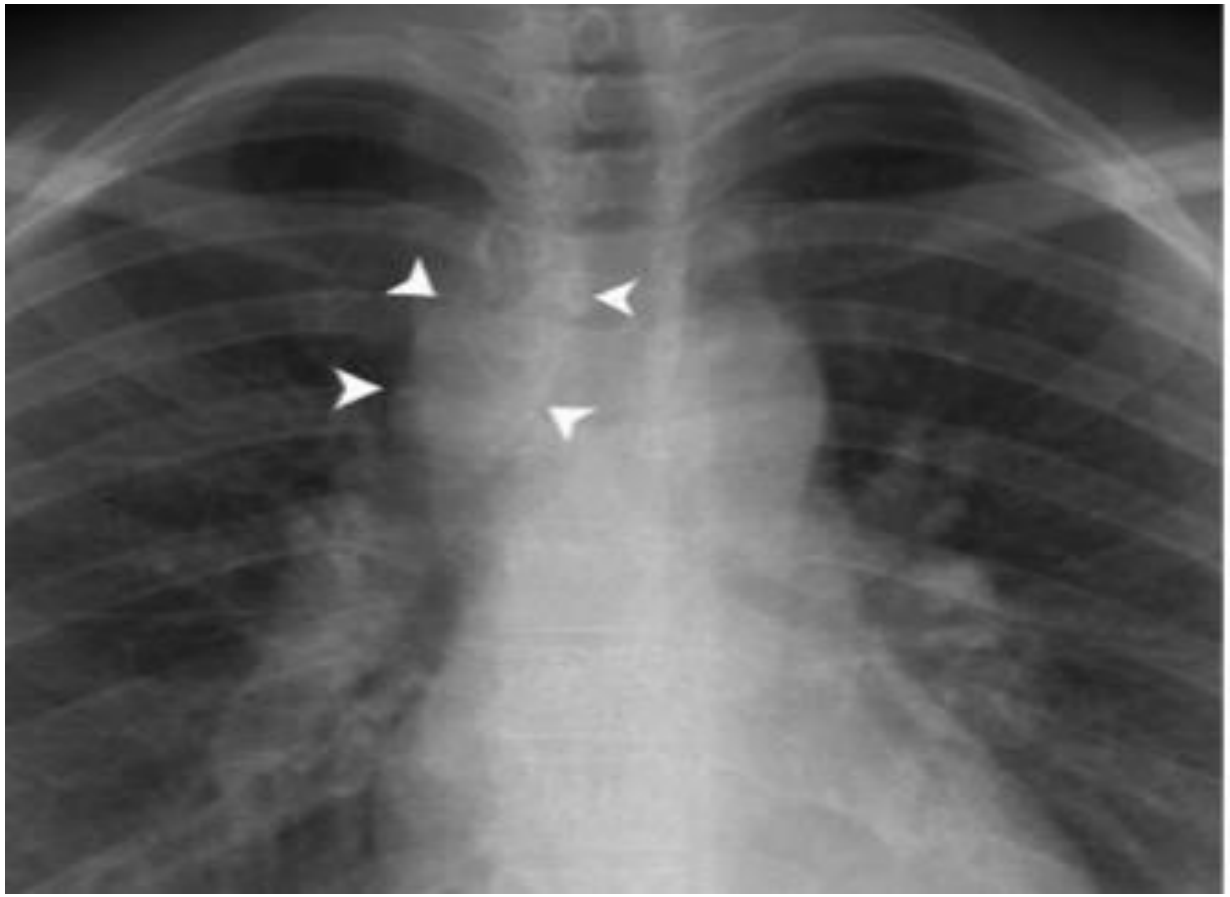

A



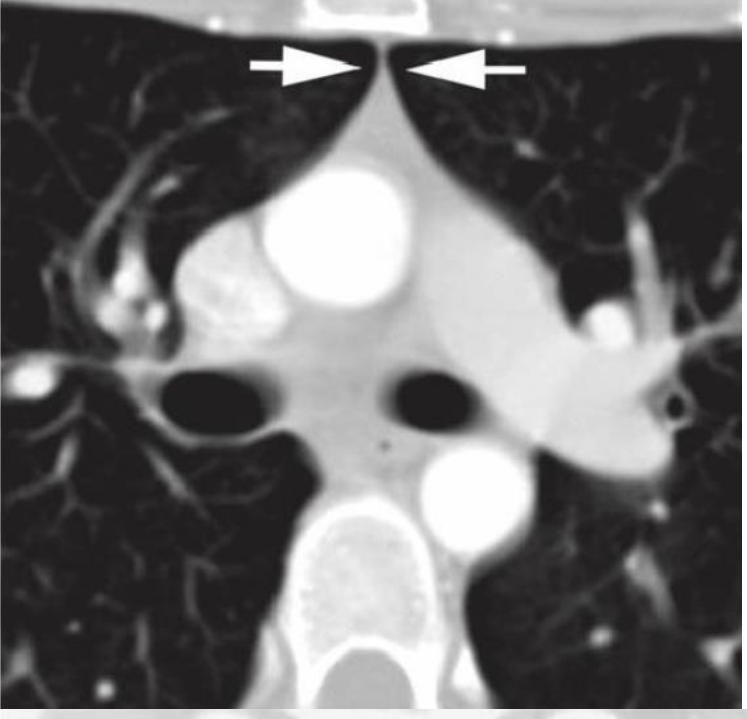
LEFT: A lung mass abutts the mediastinal surface and creates acute angles with the lung.
RIGHT: A mediastinal mass will sit under the surface of the mediastinum, creating obtuse angles with the lung

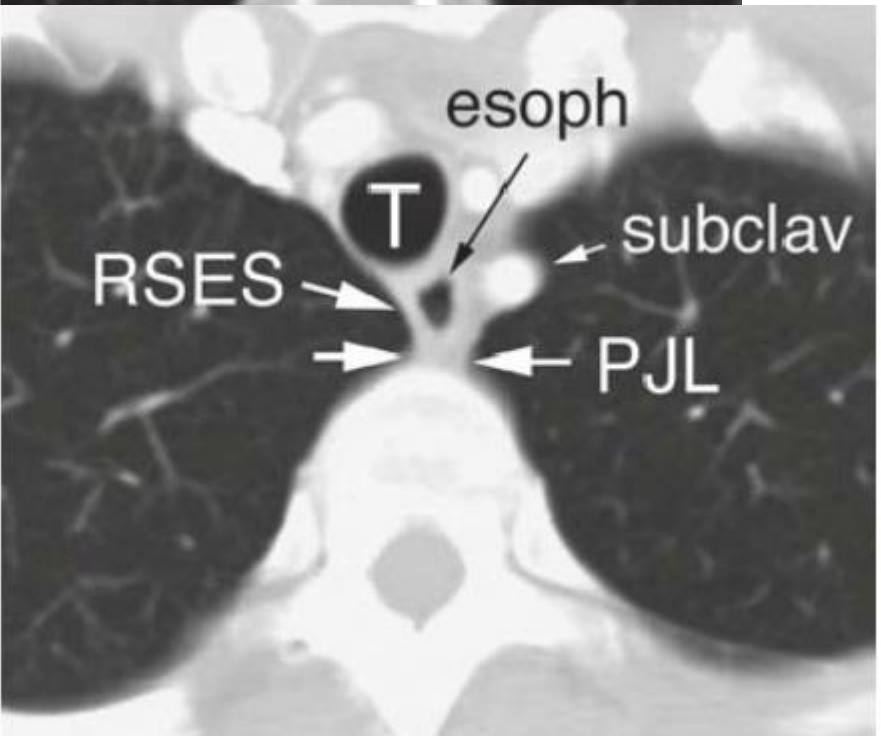
Courtesy – Radiology assisstant

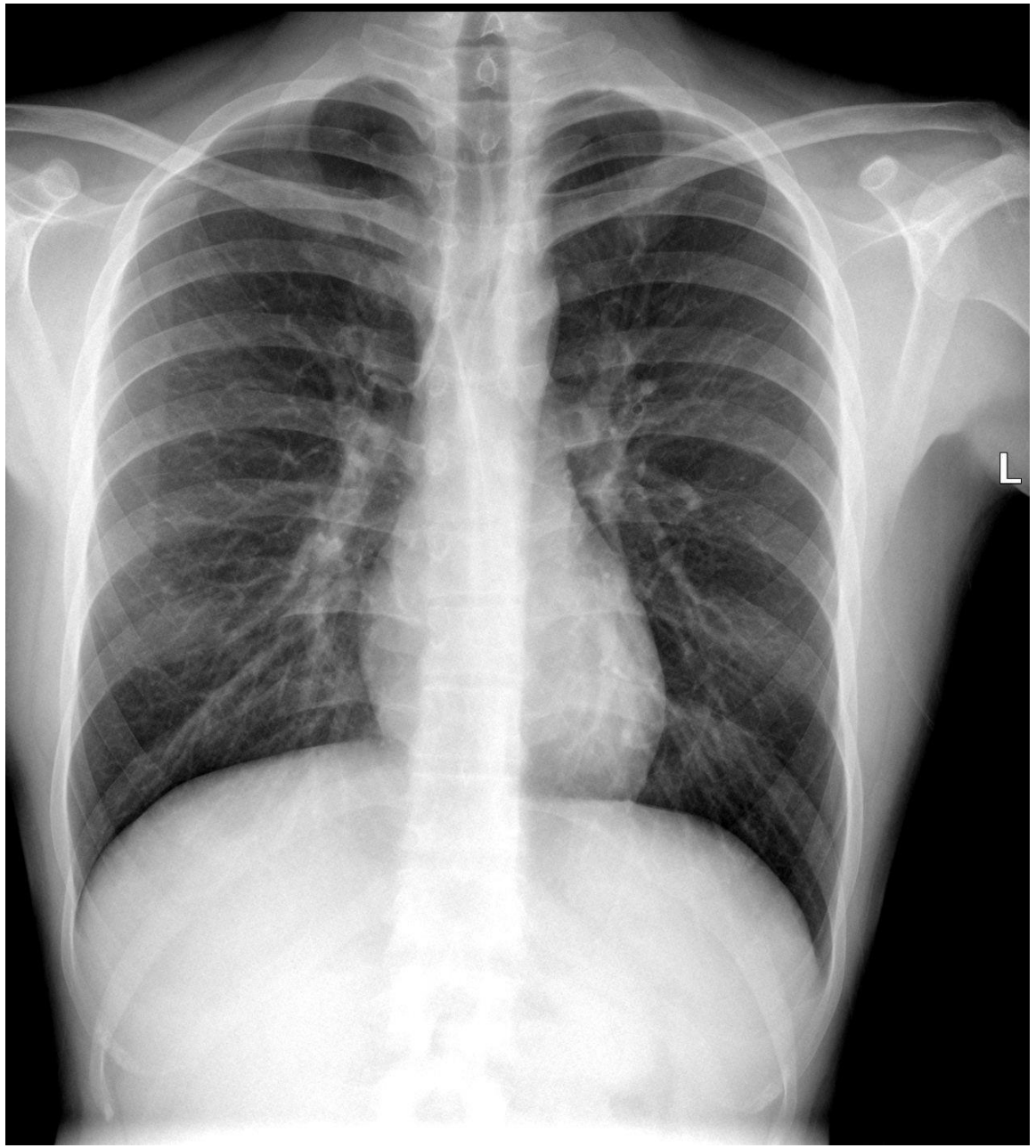
Widening of paratracheal stripe

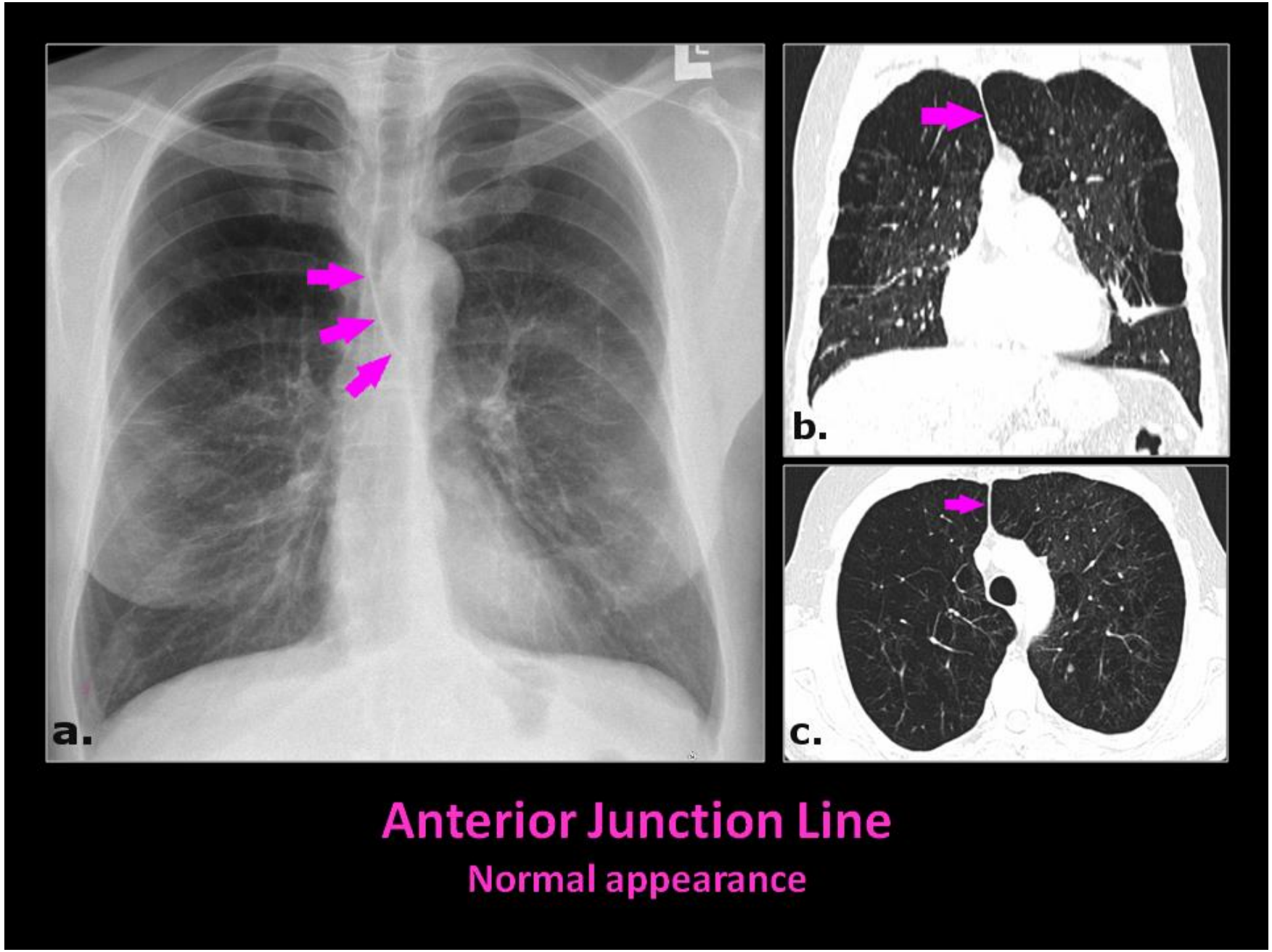


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- Normal Rt = < 5mm, Lt = <10mm.
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- Tracheal wall tumors.
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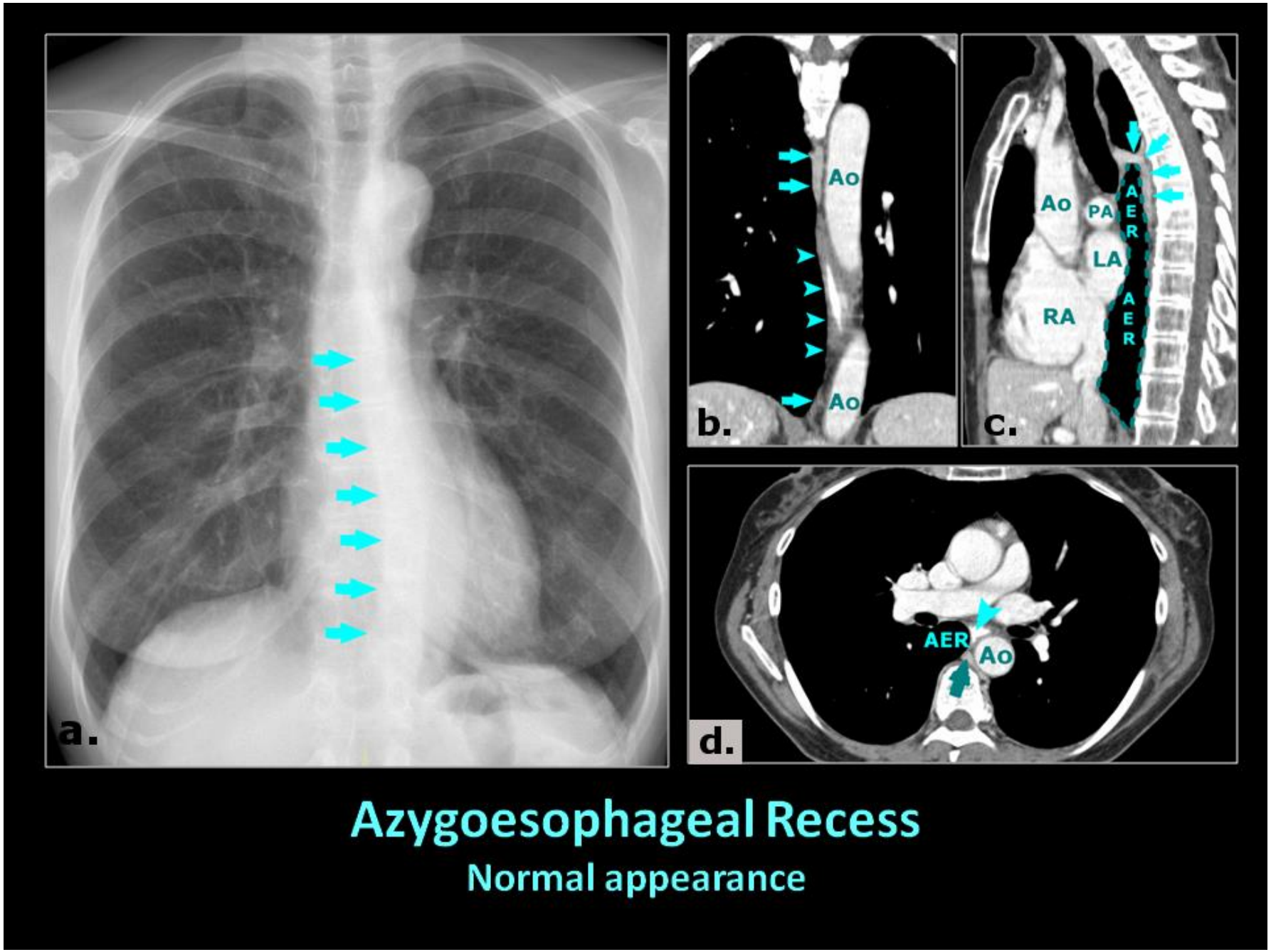

esoph
T
subclav
RSES
PJL


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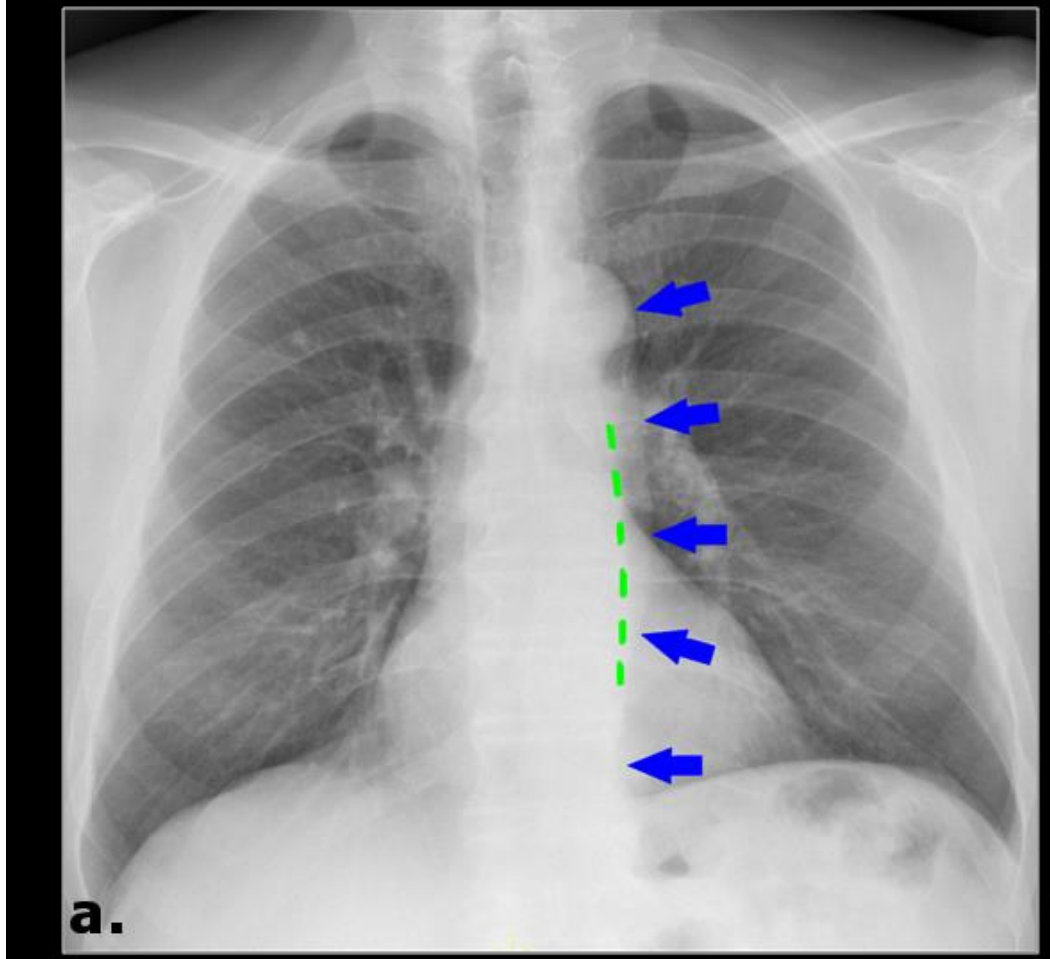
Anterior Junction Line

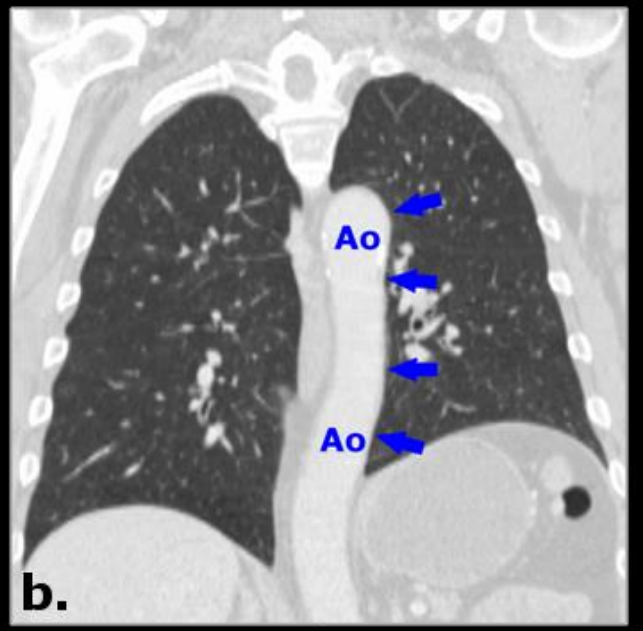
Normal appearance

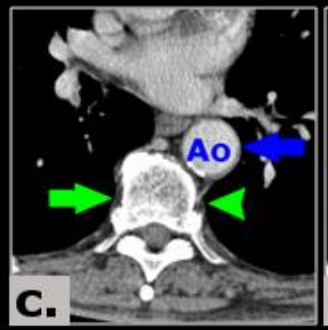


Azygoesophageal Recess

Normal appearance



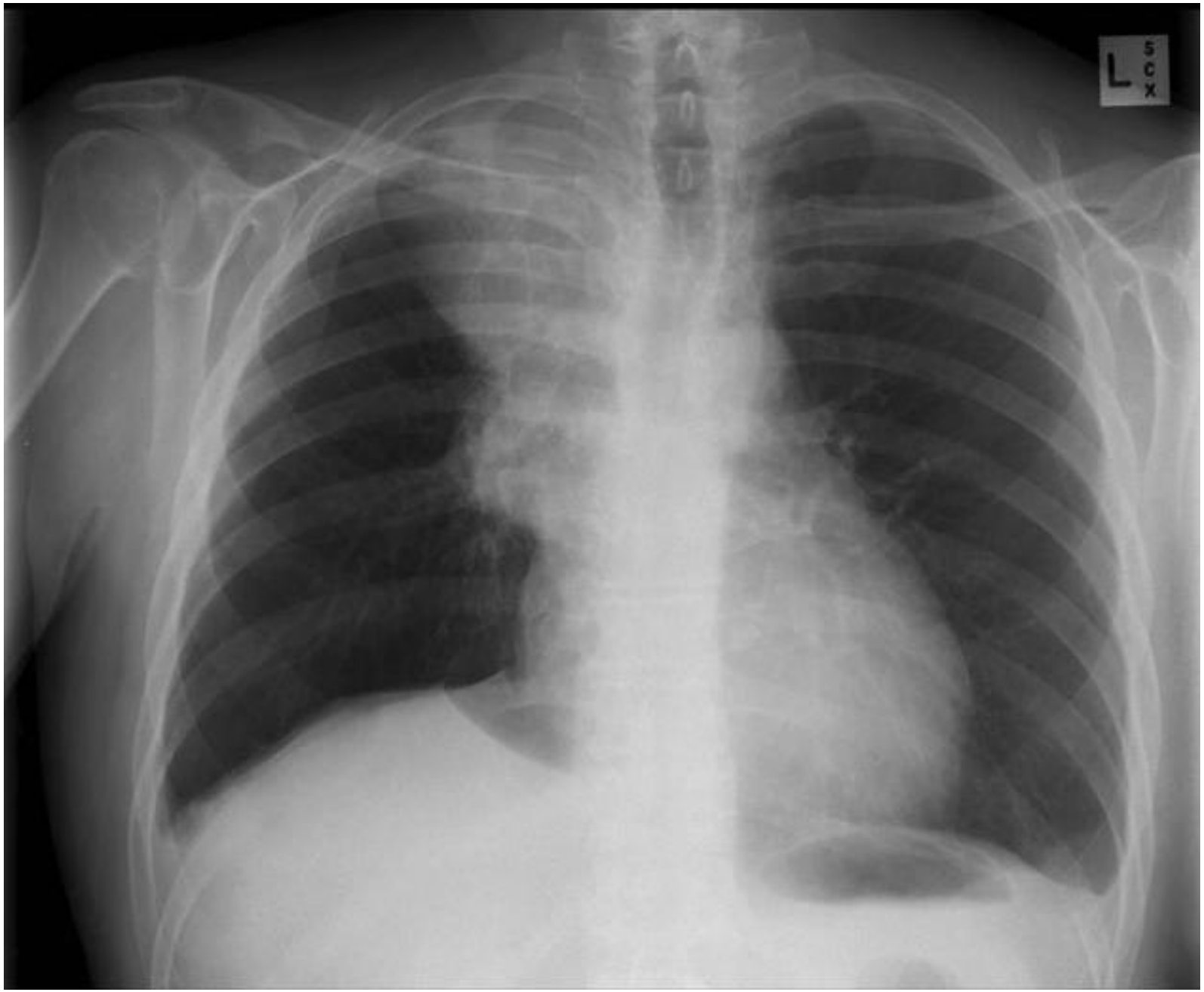


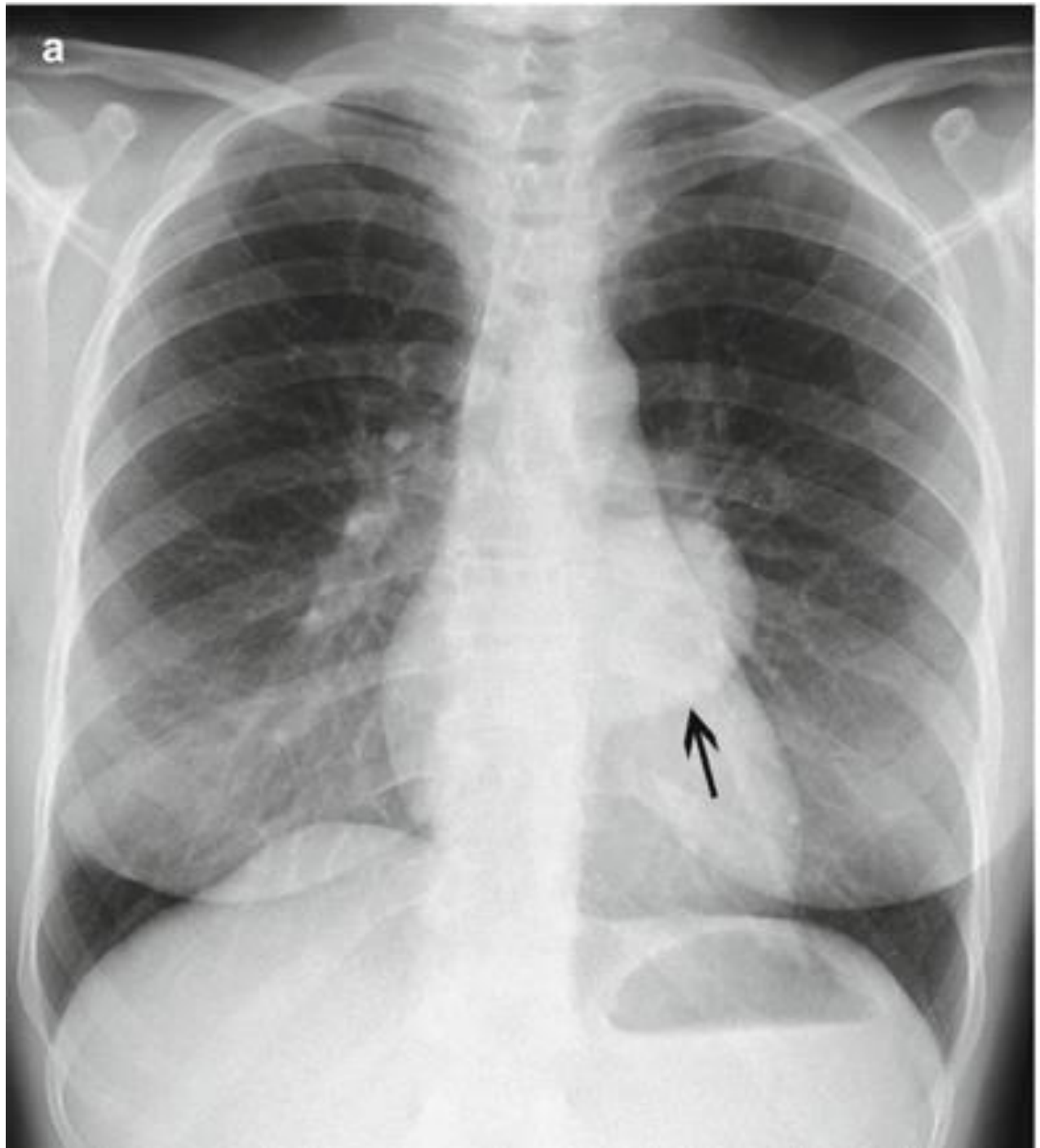


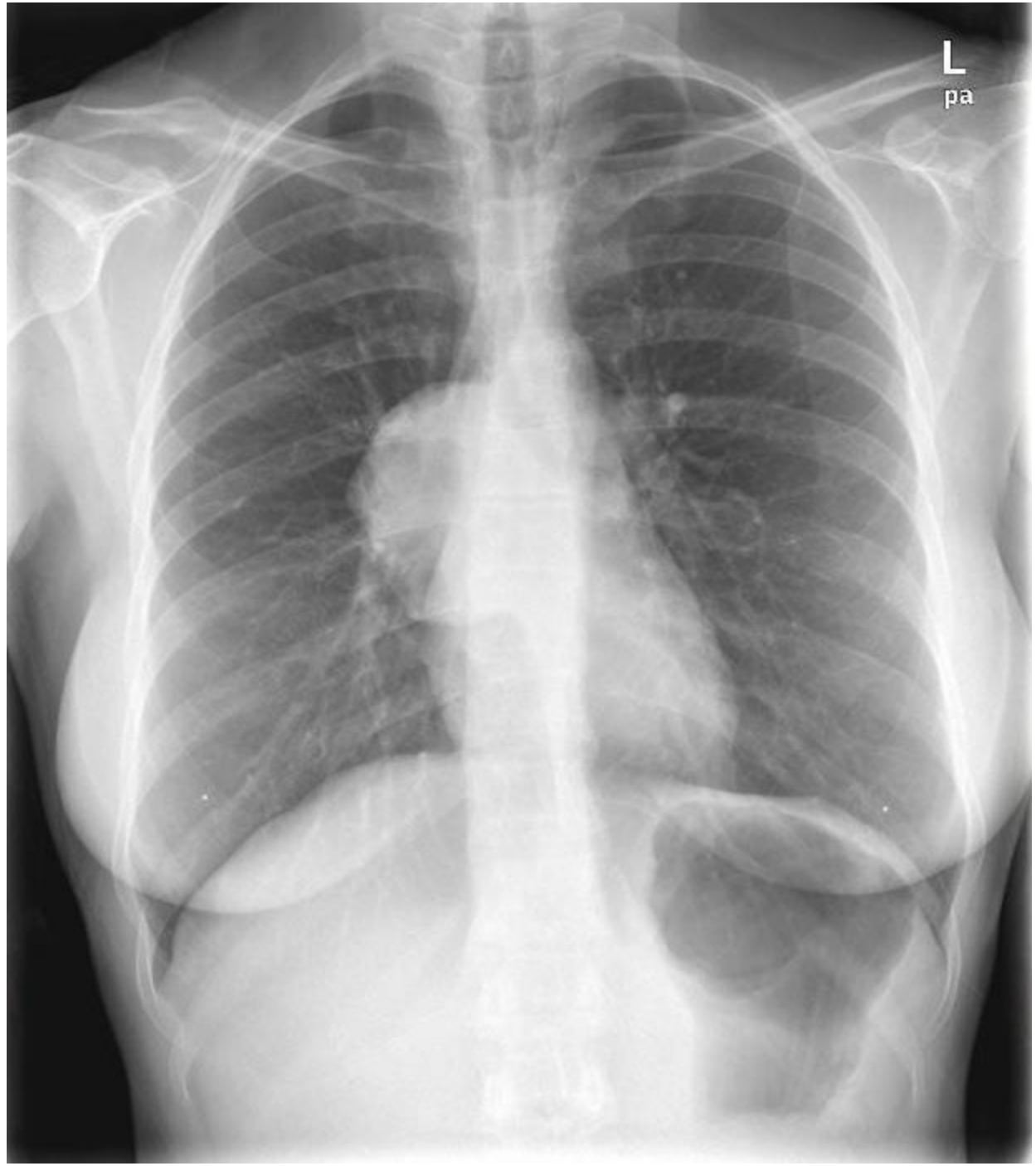


Paraortic Line

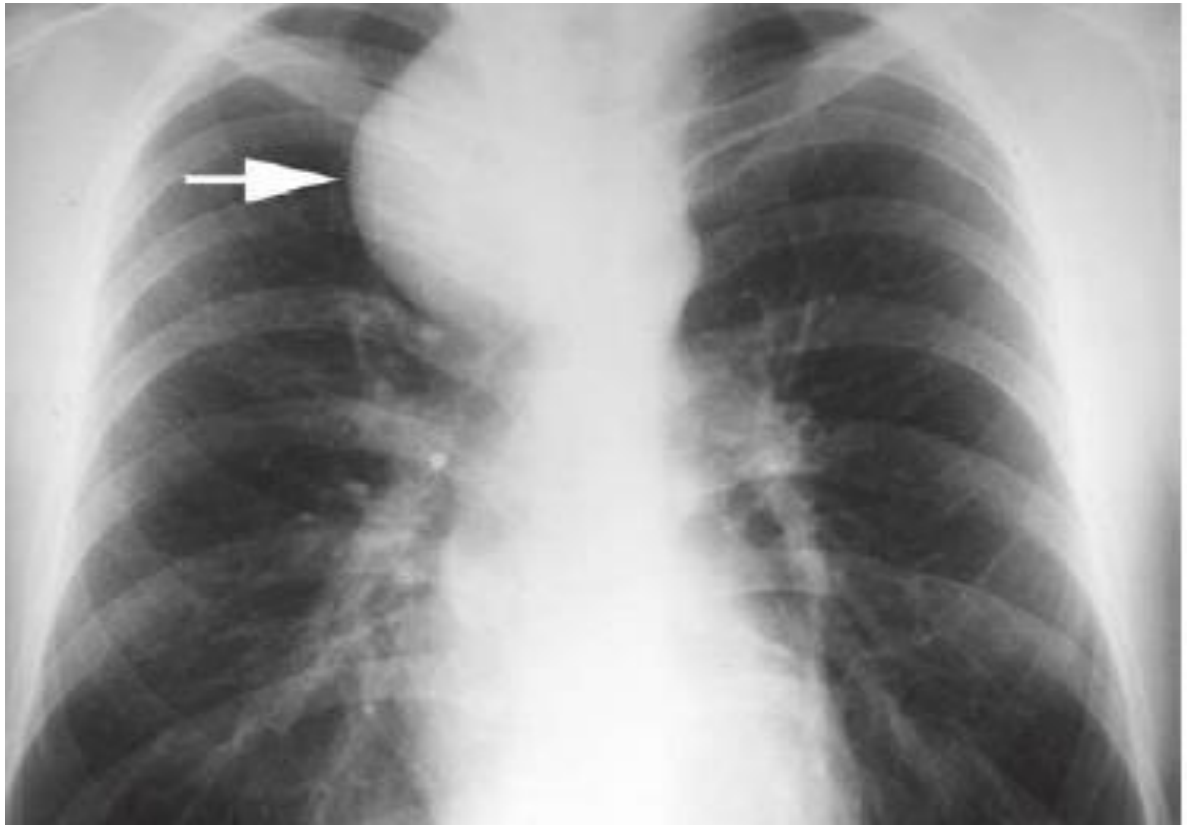
Normal appearance

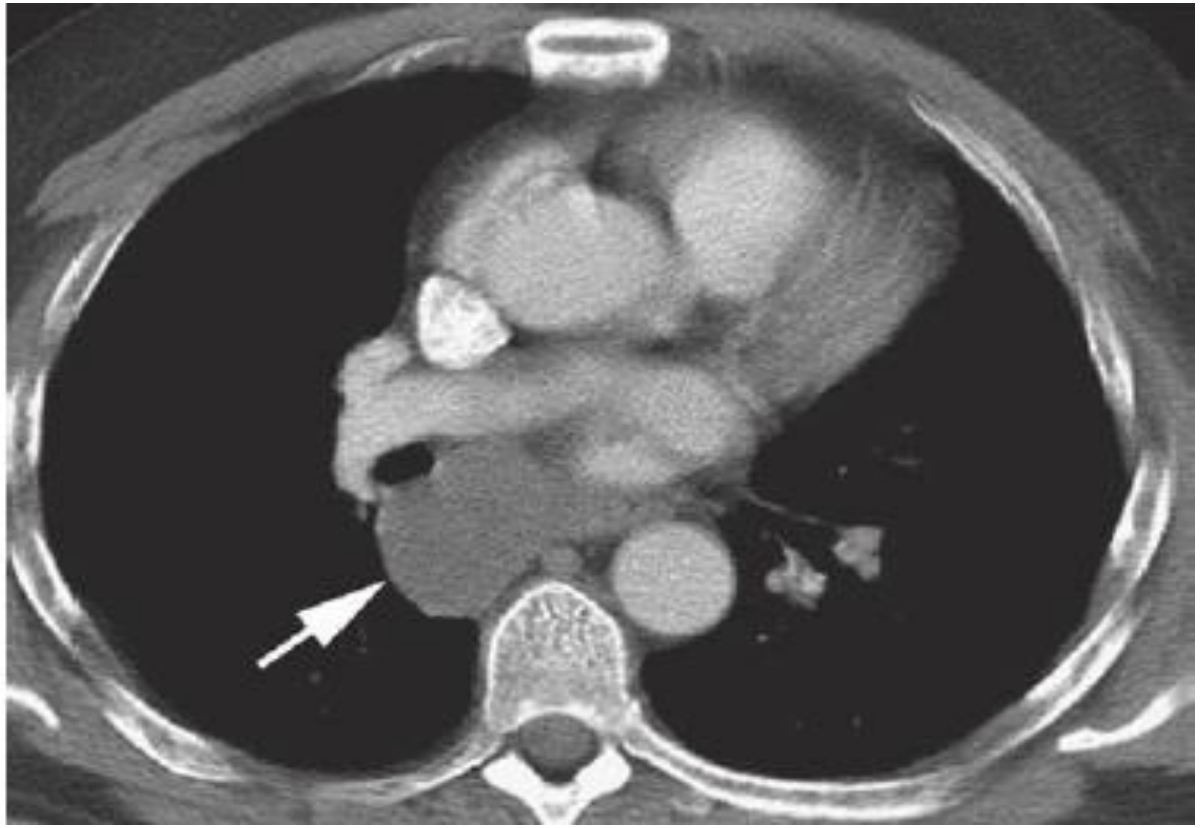

L
5
C
X


a

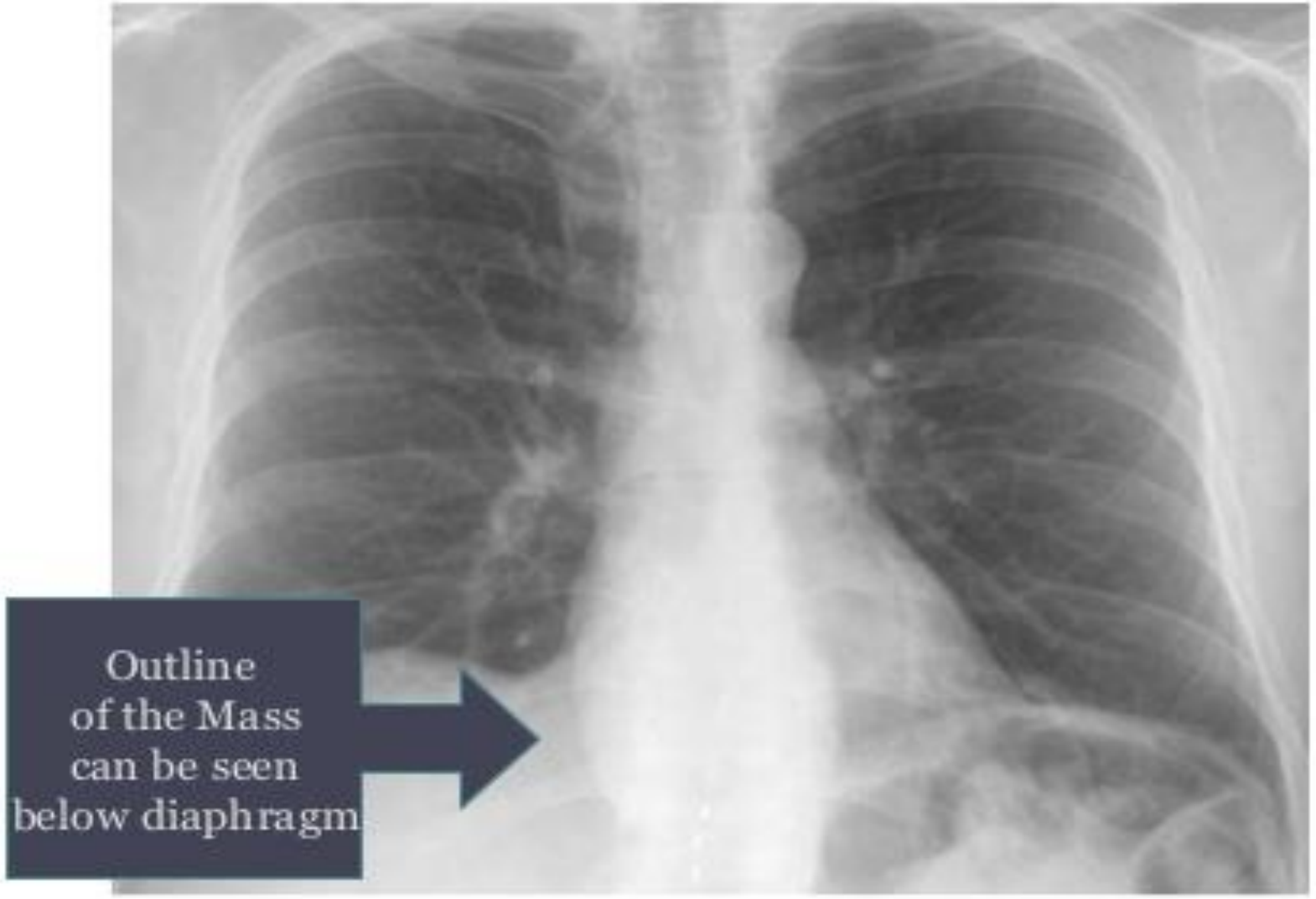

L
pa

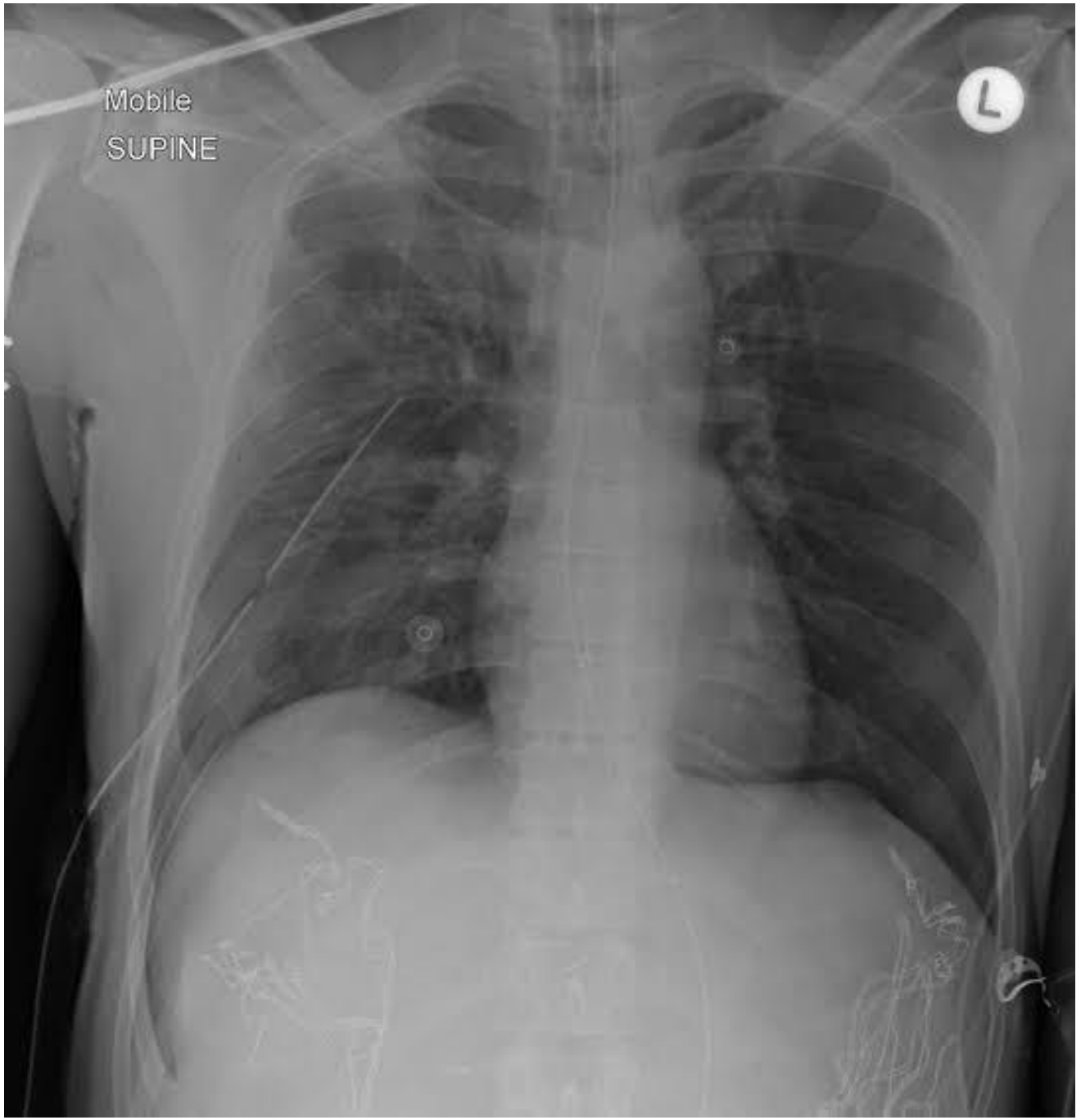
Cervicothoracic sign



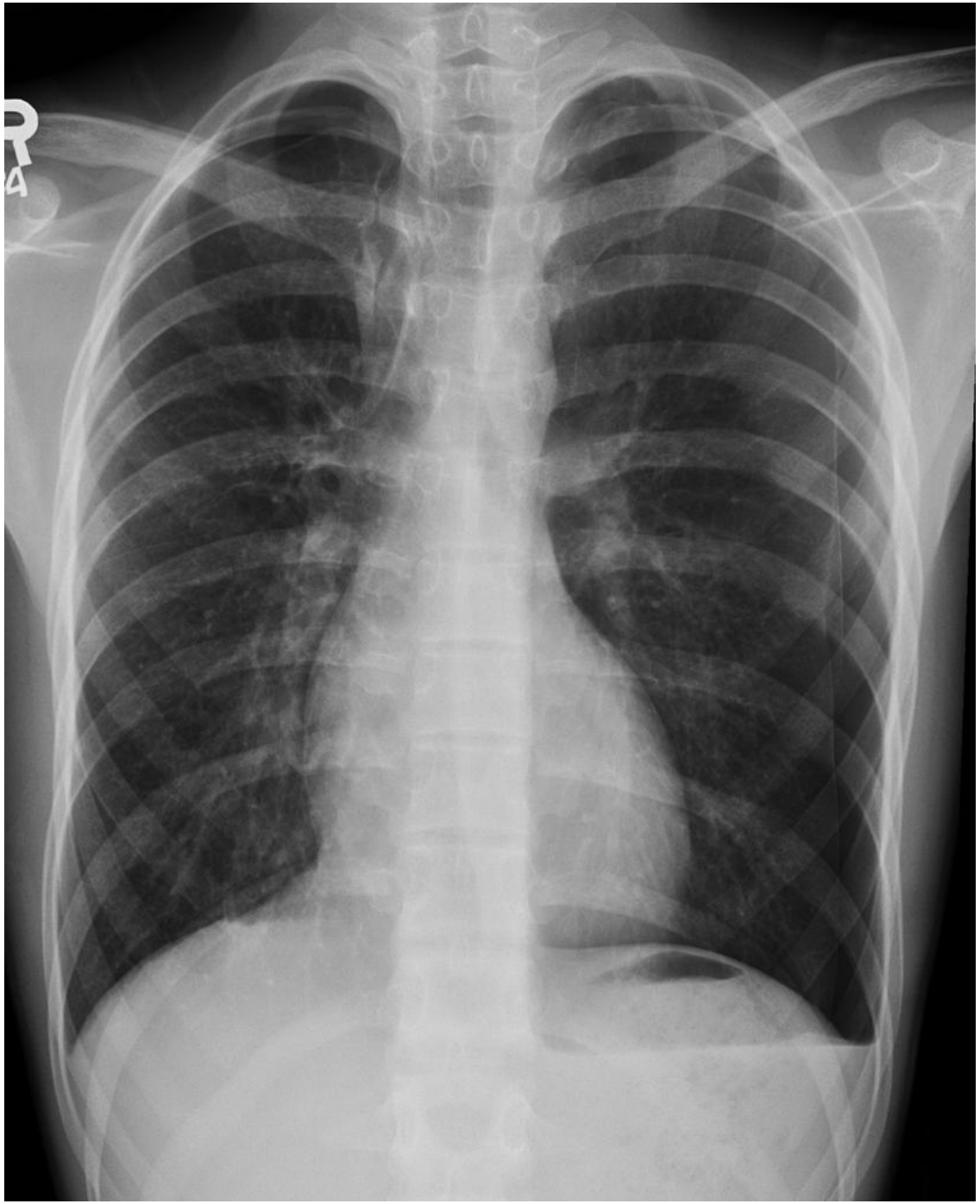


Thoracoabdominal sign:

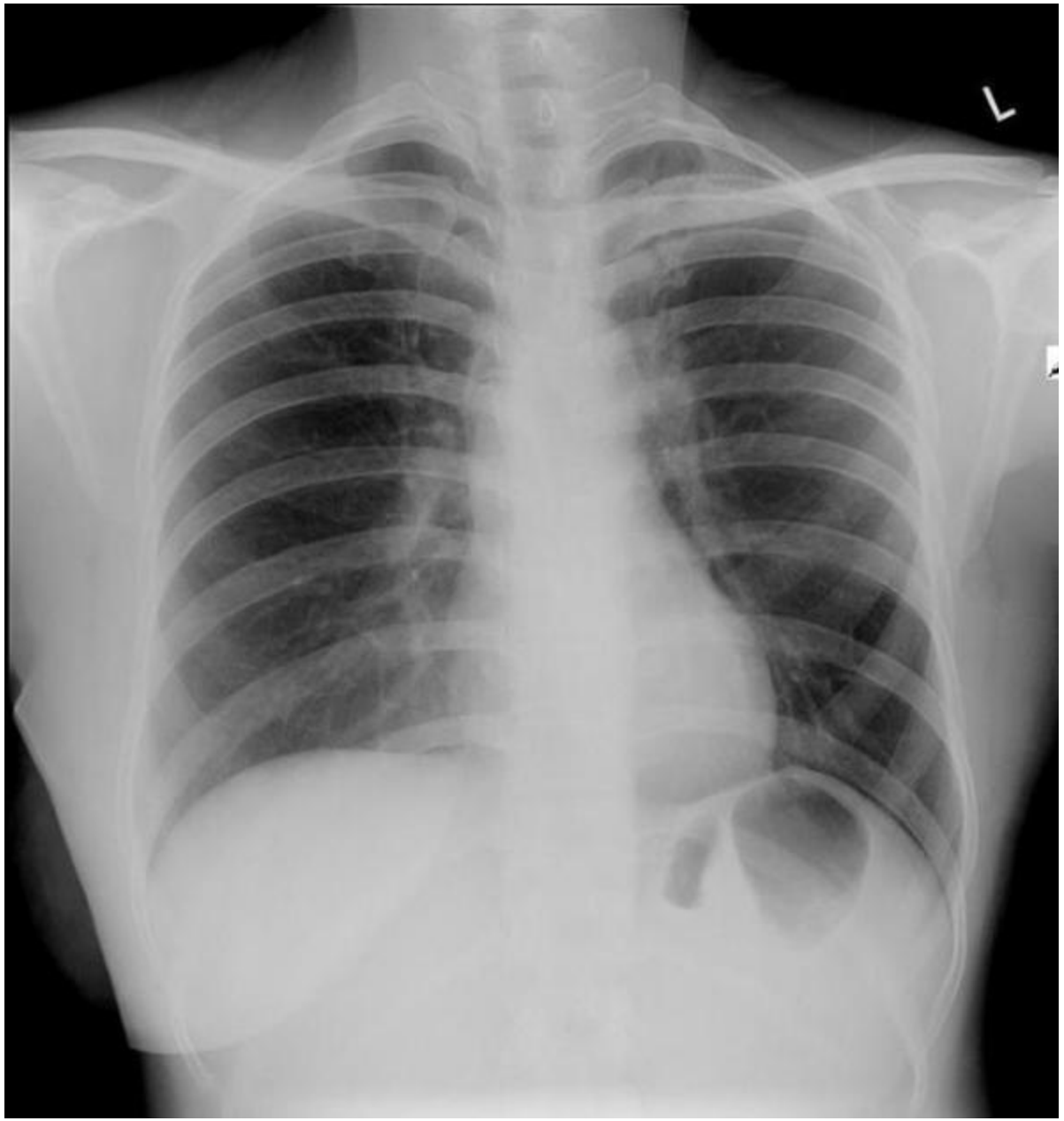




Deep Sulcus Sign



PNEUMOTHORAX



Unilateral hypertranslucency – post mastectomy

Unilateral hyper lucent lung



- Rotation
- Pneumothorax
- Mastectomy
- Polio
- Poland syndrome
- Surgical removal of pectoralis major
- obstruction,
- Emphysema
- Swyer-James syndrome
- Unilateral large bullae
- Large pulmonary embolus
- Scoliosis
- Contralateral lung increased density, e.g. supine pleural effusion.

Normal variants





- Scalloping
- Slips
- Hump

Summary
Systematic approach – Reading a Chest Radiograph

- Clinical history.
- Identification – Patient, side
- PA vs AP, Exposure factors, rotation, centering, inspiration
- Trachea.
- Hila.
- Cardiac shadow.
- Diaphragms.
- Costophrenic / Cardiophrenic angles.
- Lungs- including hidden areas.
- Soft tissues.
- Bones.



"You don't have to keep smiling
for a chest X-ray."

"Everything that we see is a shadow cast by that which we do not see."

- Martin Luther King, Jr.

Thank You